Mathematical Anatomy of Trading Systems

A Multi-Asset Long-Short Futures System

From factor geometry to an admissible trading policy

A rigorous mathematical course

Volume I

September 2026

A route through this volume

This volume builds a futures trading experiment from its cash flows. The reader needs linear algebra, multivariable calculus, elementary probability and basic analysis. Time-series dynamics, stochastic integration, filtering and control are developed when the trading problem first requires them.

Chapters 1–6 identify gains, separate common movement and choose a state. Chapters 7–13 build dynamics, inference and a predictive distribution. Chapters 14–19 turn that distribution into feasible positions, an adaptive policy, actual fills and a cash ledger. Chapter 20 tests the experiment; The closing two-page sheet makes it executable. All examples are symbolic. No profitable anomaly or empirical model fit is assumed.

How to read the mathematics

Work through the scalar example before its vector generalization. Vectors are columns and $\top$ denotes transpose. A covariance is positive semidefinite; a displayed ordinary inverse always needs the stronger conditions stated nearby. Dates are discrete until elapsed time requires continuous notation. The physical probability $\mathbb{P}$ describes actual market outcomes; conditioning always uses information already available.

The main text introduces each object when its purpose becomes visible. Sections marked *Derivation* work through the algebra behind a result. Technical appendices retain domain restrictions, exceptional cases and longer proofs. A forward reference to an appendix is permission to defer that detail, not a prerequisite for understanding the next paragraph. The notation table is a reference; no symbol in it needs to be memorized first.

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Principal notation

Symbol	Meaning and space
$(\Omega, \mathcal{F}, \mathbb{P}), \mathcal{G}_t$	Probability space and information actually available at decision t.
$F_t^{i,c}, m_i$	Quoted futures price of maturity c, cash multiplier per price unit.
$c_i(t), d_{i,t} > 0$	Predictably selected contract and positive gain-normalization scale.
$r_{t+1} \in \mathbb{R}^n$	Normalized futures gain vector; not a return on invested premium.
$q_t^-, q_t^+ \in \mathbb{Z}^n$	Contract inventories before and after trading; real relaxations are stated.
$a_t, w_t \in \mathbb{R}^n$	Filled contract change and normalized dollar exposures.
W_t, C_t, M_t	Total equity, freely available cash, posted collateral.
B, V_k, U	Factor loading matrix, orthonormal factor basis, residual basis.
$P_k = V_k V_k^\top$	Euclidean orthogonal projector onto the factor subspace.
$x_t \in \mathbb{R}^d, y_t$	Latent statistical state and observed state proxy.
A, Q, R	State-transition matrix, state-noise covariance, observation-noise covariance.
ϑ, Θ	Unknown statistical parameter and its admissible parameter space.
b_t	Posterior law of latent state and unknown parameters given $\mathcal{G}_t$.
μ_t, Σ_t	Conditional mean and covariance of next-period normalized gains.
$\gamma > 0, \beta$	Risk-aversion coefficient and dynamic discount factor; distinct quantities.
$C_t, \mathcal{A}(s)$	Portfolio feasible set and state-dependent feasible actions.
J_t, V_t	One-period decision criterion and dynamic continuation value.
π, K^π	Trading policy and its induced state transition kernel.
$L, \mathrm{ES}_\alpha(L)$	Loss random variable and upper-tail expected shortfall.

Chapter 1

What a futures position earns

What problem are we solving?

Before forecasting a price, we must know which price changes actually put money in the account. A futures quote looks like an asset price, but opening a futures contract does not purchase that quoted notional. We begin with one contract and one settlement interval.

1.1 One position, one interval

Write F_t for today's settlement quote of a particular contract and F_{t+1} for its next settlement quote. The subscript labels dates, not powers. Let $m > 0$ be the contract multiplier: a one-unit quote change transfers m cash units per contract. If we hold q contracts, the settlement gain is

$$G_{t+1} = qm(F_{t+1} - F_t). \tag{1.1}$$

Here $q > 0$ means long and $q < 0$ means short. Reversing the sign reverses the gain. This is an accounting identity for the contract held across the interval; it makes no claim about which sign will earn money on average.

For a long position, a rise in the settlement quote credits cash through variation margin; a fall debits cash. The initial performance bond is collateral supporting these obligations. It is not the purchase price of a security [5]. Consequently qmF_t, the quoted notional, does not enter the initial cash ledger as a premium payment. Negative quotes do not invalidate (1.1).

1.2 From a gain to a forecast question

If we choose q today, only the next quote is unknown in (1.1). The simplest forecasting question is therefore: what is the distribution of $m(F_{t+1} - F_t)$, given what we know today? An expected price change can suggest a direction. Its uncertainty and the costs of acquiring the position will later determine whether that direction is useful. For now, keep the distinction between the gain identity and a model for its law.

1.3 A first cash check

Let W_t be account equity, let $K_t \geq 0$ be the cost of changing the position, and temporarily ignore interest. With no deposit or withdrawal,

$$W_{t+1} = W_t - K_t + qm(F_{t+1} - F_t).$$

Cash posted as collateral remains part of W_t. Later we separate available cash from collateral because they have different uses and may earn different rates. That refinement must preserve this simple conservation principle.

Key idea

The object to forecast is a gain earned by an identified holding. A transformation of a chart is useful only after it has been connected to such a holding.

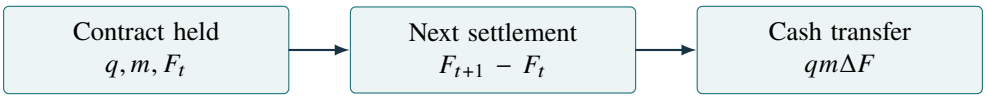


Figure 1.1: The first arrow identifies the same contract at both dates; the second converts its quote change into cash.

What we gained

- A signed contract position and its exact settlement gain.
- A distinction between notional, collateral and profit.
- A cash identity against which later models can be checked.

Where this goes next

With several contract families, quote changes have different units and maturities. We need a vector whose coordinates can be modeled together while retaining the cash conversion.

Chapter 2

The tradable gain vector

What problem are we solving?

A covariance matrix cannot sensibly compare a raw quote change in one contract with a differently scaled quote change in another. We introduce a positive normalization, then check that rolling maturities does not create fictitious gains.

2.1 Information means what is known now

Throughout the book, $\mathcal{G}_t$ denotes information actually available at date t. It includes published quotes and received account notices. The notation $\mathbb{E}[Y \mid \mathcal{G}_t]$ means the average of an uncertain quantity Y conditional on that information. In a finite example, if today's observation is event A, this average is $\sum_y y\,\mathbb{P}(Y = y, A)/\mathbb{P}(A)$, for $\mathbb{P}(A) > 0$. The symbol $\mathbb{P}$ names the probability law for actual market outcomes.

A choice is *predictable* here if it is made before the price increment to which it applies. Tomorrow's settlement cannot choose today's holding or normalization. Appendix A gives the precise information and conditional-expectation definitions when observations are not finite-valued.

2.2 Choosing units before computing statistics

For contract family i, divide cash gain by a positive cash scale $d_{i,t}$ known today. This makes a dimensionless coordinate. Multiplying back by the same scale must recover cash exactly. A normalization is a choice of units, not a source of economic gain.

2.3 Contracts and normalized gains

For each family $i \in \{1, \dots, n\}$, let $F_t^{i,c} \in \mathbb{R}$ denote the settlement quote of maturity c, and let $m_i > 0$ convert one price unit into cash. A long contract held across a settlement interval earns

$$\Delta H_{t+1}^{i,c} = m_i (F_{t+1}^{i,c} - F_t^{i,c}). \tag{2.1}$$

It does not require paying $m_i F_t^{i,c}$ as a purchase premium. This is why the ratio of a price change to posted margin is not an intrinsic return of the underlying contract. Daily marking and performance bonds are distinct mechanisms [5].

Let $c_i(t)$ be the contract chosen at t, known in $\mathcal{G}_t$, with no delivery during the holding interval. Select a strictly positive, predictable cash scale $d_{i,t}$, with both $d_{i,t}$ and $d_{i,t}^{-1}$ bounded on each admissible research block. Define

$$r_{i,t+1} = \frac{m_i(F_{t+1}^{i,c_i(t)} - F_t^{i,c_i(t)})}{d_{i,t}}, \qquad D_t = \operatorname{diag}(d_{1,t}, \dots, d_{n,t}). \tag{2.2}$$

Thus r_{t+1} is dimensionless and $q^\top D_t r_{t+1}$ is cash. A positive scale avoids undefined log returns and sign inversions when a futures quote is zero or negative. Where prices are strictly positive, choosing $d_{i,t} = m_i F_t^{i,c_i(t)}$ recovers the simple futures price return, but that is an optional convention.

Signed quoted notional $m_i q_i F_t^{i,c}$, gross reference exposure $|q_i| d_{i,t}$, and required margin are different objects. If all accounts use a common currency, the construction applies directly. For contracts settled in another currency, include the local cash account and exchange rate X_t^i in the state: converted gains are $X_{t+1}^i \Delta H_{t+1}^{i,c}$, and retained local cash also earns exchange-rate P&L. The main system either trades base-currency-settled contracts or explicitly hedges and accounts for these additional cash risks.

Reading the vector identity

The matrix D_t places the scales on its diagonal. Thus $D_t r_{t+1}$ is the vector of cash gains from one contract of each family, and $q^\top D_t r_{t+1}$ sums the gains of a portfolio. When later chapters write $w^\top r$, w is an exposure in these chosen units. A change of units must change both the gain coordinate and its exposure.

2.4 Spot, basis, carry and rolling

The remaining difficulty is contract identity. A front-contract chart can switch from one maturity to another. The following calculation distinguishes movement of a held contract from movement caused by relabeling the chart.

Let S_t^i denote a comparable spot quote when one exists, and let $b_t^{i,c} = F_t^{i,c} - S_t^i$ be the basis. Holding the same maturity gives the identity

$$\Delta F_{t+1}^{i,c} = \Delta S_{t+1}^i + \Delta b_{t+1}^{i,c}. \tag{2.3}$$

Spot total return also includes income or storage costs of owning spot. Futures excess gain is settlement variation; collateral return arises from the separate cash account. Neither is a dividend received by a futures holder.

For a storable commodity in a frictionless carry model, a forward quote may satisfy $F(t,T) = S_t \exp\{\int_t^T (r_u + u_u - y_u)\,\mathrm{d}u\}$, where r is financing, u storage and y convenience yield. This formula requires the relevant replication assumptions; some assets have no accessible spot storage strategy. Futures and forward quotes can differ with stochastic interest rates because settlement occurs at different times. Accordingly our ledger needs observed futures prices, not universal validity of a carry formula.

Write $\mathfrak{F}(t,\tau) = F(t, t+\tau)$ for a smooth maturity-curve representation. Along a fixed delivery date T, remaining maturity decreases at rate one. Its gain separates, under a declared curve-motion convention, into a fixed-τ curve change and $-\partial_\tau \mathfrak{F}\,\mathrm{d}t$. This latter term is roll-down carry, not money received when a contract is exchanged.

If an artificial series $A_t = F_t^{i,c_i(t)}$ switches maturity, then

$$A_{t+1} - A_t = \underbrace{F_{t+1}^{i,c_i(t)} - F_t^{i,c_i(t)}}_{\text{gain of the contract actually held}} + \underbrace{F_{t+1}^{i,c_i(t+1)} - F_{t+1}^{i,c_i(t)}}_{\text{splice gap}}. \tag{2.4}$$

The gap is not a settlement gain. A roll closes one zero-marked contract and opens another, incurring execution costs and changing future curve exposure. A continuous adjusted quote can be useful for a statistical hypothesis, but only (2.2) earns cash. Our base policy rolls before delivery under a predictable eligibility rule and includes both legs' costs.

What we gained

- A normalized vector of gains with an exact cash conversion.
- An information rule that forbids using future observations.
- A separation of settlement gain, basis movement, roll-down and splice gaps.

Where this goes next

The coordinates can now be compared. We next ask whether much of their movement is shared, and which linear combinations remove that shared movement.

Chapter 3

Common movement and relative movement

What problem are we solving?

Several contracts may react to the same economic shock. A portfolio intended to express a relative forecast can accidentally become a large bet on that common shock. We first separate the two kinds of movement in three assets.

3.1 Three assets and one common shock

Suppose centered normalized gains have the form

$$r_1 = \beta_1 f + e_1, \qquad r_2 = \beta_2 f + e_2, \qquad r_3 = \beta_3 f + e_3.$$

The scalar f is a common shock; β_i tells how strongly contract i responds; e_i is what remains in that contract. For exposures w_1, w_2, w_3, direct multiplication gives

$$\sum_i w_i r_i = (\beta_1 w_1 + \beta_2 w_2 + \beta_3 w_3) f + \sum_i w_i e_i.$$

The coefficient of f is therefore the quantity to cancel. If $\beta_3 \neq 0$, every neutral exposure can be written

$$w = u \begin{pmatrix} 1 \\ 0 \\ -\beta_1/\beta_3 \end{pmatrix} + v \begin{pmatrix} 0 \\ 1 \\ -\beta_2/\beta_3 \end{pmatrix}, \qquad u, v \in \mathbb{R}.$$

The two free numbers specify bets still available after removing the common shock. Geometrically they parameterize a plane perpendicular to $\beta = (\beta_1, \beta_2, \beta_3)^\top$. This plane is the nullspace of the row $\beta^\top$. It contains many portfolios; the cancellation condition alone does not choose a profitable one.

3.2 Decomposing a realized vector

If $\beta \neq 0$, put $v = \beta/\|\beta\|$. The closest point to a realized gain vector r on the factor line is $v(v^\top r)$: minimize $\|r - av\|^2$ over the scalar a, differentiate, and obtain $a = v^\top r$. The remaining vector is perpendicular to v. This calculation estimates a common component of a realized observation. It does not yet forecast the next observation.

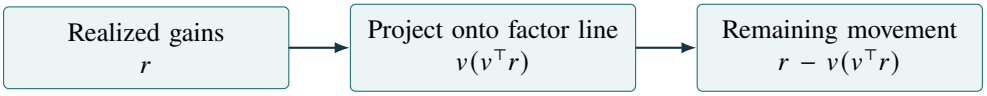


Figure 3.1: Projection answers a cross-sectional reconstruction question. Predicting the residual over time is a separate task.

3.3 A factor representation

We can now replace one common shock by several. The columns of the loading matrix play the role of the vector of coefficients in the three-asset example.

For the population geometry, let $m_r = \mathbb{E}[r_t]$ be the unknown common mean in the maintained stationary block, and let the truly centered vector $z_t = r_t - m_r \in \mathbb{R}^n$ satisfy

$$z_t = Bf_t + \varepsilon_t, \qquad B \in \mathbb{R}^{n\times k}, \quad f_t \in \mathbb{R}^k, \quad \varepsilon_t \in \mathbb{R}^n. \tag{3.1}$$

The population PCA derivation below uses this zero-mean z_t. In estimation, replace m_r by a training estimate $\bar{r}$ fixed before the evaluation block. Those future deviations need not have zero physical mean; the discrepancy is intercept/forecast uncertainty, not silently zero. The exact coordinate decompositions later use $r_t - \bar{r}$, while reconstruction identities involving population covariance here use $r_t - m_r$.

The factors may be observed economic returns or latent coordinates. If latent, $Bf = (BR)(R^{-1}f)$ for every invertible R, so loadings and factor coordinates are not separately identifiable without normalization. An orthonormal B removes scale but still permits rotations.

With finite second moments, define $\Sigma_f = \mathrm{Cov}(f)$, $\Sigma_\varepsilon = \mathrm{Cov}(\varepsilon)$ and $C_{f\varepsilon} = \mathrm{Cov}(f, \varepsilon) \in \mathbb{R}^{k\times n}$. Expanding the centered outer product gives

$$\Sigma_r = B\Sigma_f B^\top + \Sigma_\varepsilon + BC_{f\varepsilon} + C_{f\varepsilon}^\top B^\top. \tag{3.2}$$

The familiar two-term formula requires $C_{f\varepsilon} = 0$. Unconditional orthogonality in a training window does not imply zero conditional cross covariance in tomorrow's forecast.

A normalized dollar exposure w earns $w^\top r$. Its factor coefficient is $B^\top w$. Hence factor neutrality means $B^\top w = 0$, not merely $\mathbf{1}^\top w = 0$, and does not remove omitted factors, nonlinear effects or estimation error. If factors are extracted from scaled gains $z = L^{-1}(r - \bar{r})$, the return loading is LB and neutrality is $B^\top L^\top w = 0$.

What we gained

- A worked neutral portfolio with two remaining degrees of freedom.
- Projection onto a common direction and the perpendicular residual.
- The general loading map and the covariance cross terms it produces.

Where this goes next

So far the common direction was supplied. PCA chooses directions from covariance by an explicit reconstruction criterion; we derive that criterion next.

Chapter 4

PCA and factor geometry

What problem are we solving?

We want a few directions that describe much of the joint movement without guessing each loading. Linear algebra supplies an optimization problem, but its objective must be understood before its eigenvectors are interpreted.

4.1 What should a good direction do?

Imagine replacing every centered observation z by its shadow on one line. A useful line leaves a small average squared residual. The Pythagorean identity $\|z\|^2 = (v^\top z)^2 + \|z - vv^\top z\|^2$, for $\|v\| = 1$, says this is equivalent to making the shadow's variance large. We already know how to differentiate quadratic forms; the next derivation turns that fact into PCA.

4.2 PCA begins as a variational problem

Let $\Sigma = \mathrm{Cov}(z) \in \mathbb{S}^n_+$. The set $\mathbb{S}^n_+$ consists of symmetric positive semidefinite matrices: $u^\top \Sigma u \geq 0$ for every vector u. This holds for a covariance because that quadratic form is a variance. We write $\Sigma \succeq 0$ for the same property. A unit direction v produces scalar coordinate $v^\top z$, whose variance is $v^\top \Sigma v$. The most variable direction solves

$$\max_{v \in \mathbb{R}^n} v^\top \Sigma v \quad \text{subject to } v^\top v = 1. \tag{4.1}$$

Derivation: the maximizing direction

Its Lagrangian is $v^\top \Sigma v - \lambda(v^\top v - 1)$. Differentiation gives $2\Sigma v - 2\lambda v = 0$, so every stationary direction is an eigenvector. The real symmetric spectral theorem gives an orthonormal eigenbasis u_i with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$. Writing $v = \sum_i c_i u_i$ yields $v^\top \Sigma v = \sum_i \lambda_i c_i^2 \leq \lambda_1 \sum_i c_i^2 = \lambda_1$. Thus an eigenvector for the largest eigenvalue is a global maximizer.

Requiring $v^\top v_j = 0$ for already selected directions restricts this expansion to the remaining eigenspaces, giving successive maxima $\lambda_2, \ldots, \lambda_k$. For $V_k = (v_1, \ldots, v_k)$,

$$V_k \in \operatorname*{arg\,max}_{V^\top V = I_k} \mathrm{tr}(V^\top \Sigma V), \qquad P_k = V_k V_k^\top, \quad P_k^\perp = I_n - P_k. \tag{4.2}$$

Indeed $P_k^2 = P_k = P_k^\top$, and

$$\mathbb{E}\|z - P_k z\|^2 = \mathbb{E}[z^\top (I - P_k) z] \tag{4.3}$$
$$= \operatorname{tr}(\Sigma) - \operatorname{tr}(V_k^\top \Sigma V_k). \tag{4.4}$$

Maximum captured variance and minimum mean squared Euclidean reconstruction error are the same optimization problem. Neither objective maximizes predictive return.

For a centered observation matrix $Z \in \mathbb{R}^{N_{\text{tr}} \times n}$, with rows equal to historical $z_t^\top$, a singular value decomposition is $Z = U_Z D_Z V_Z^\top$. Then $\widehat{\Sigma} = Z^\top Z/(N_{\text{tr}} - 1) = V_Z D_Z^2 V_Z^\top/(N_{\text{tr}} - 1)$. Right singular vectors solve empirical PCA; squared singular values divided by $N_{\text{tr}} - 1$ are covariance eigenvalues. This identity also exposes rank loss when $N_{\text{tr}} - 1 < n$.

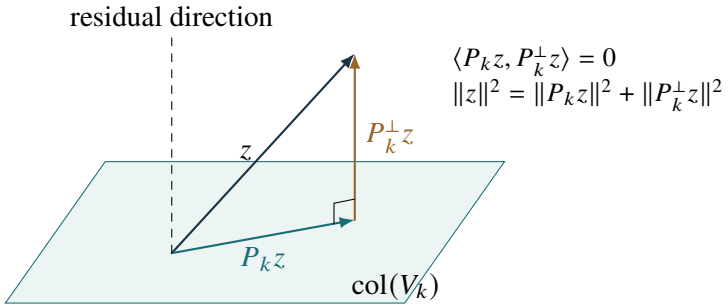


Figure 4.1: Orthogonal factor and residual components. The sketch shows one residual direction; the residual subspace generally has dimension $n - k$.

4.3 The metric is a modeling choice

Euclidean distance treats equal coordinate changes equally. A different scaling changes which reconstruction errors we penalize. We therefore derive the weighted projection instead of assuming it has the same economic meaning.

The notation $H \succ 0$, or $H \in \mathbb{S}_{++}^n$, means a symmetric matrix with $u^\top H u > 0$ for every nonzero u. It defines a squared length and has an ordinary inverse. For a positive definite metric $H \in \mathbb{S}_{++}^n$, projecting z onto the columns of a full-column-rank B minimizes $(z - Bf)^\top H (z - Bf)$. Differentiating gives

$$\widehat{f} = (B^\top H B)^{-1} B^\top H z, \qquad P_H = B(B^\top H B)^{-1} B^\top H. \tag{4.5}$$

The inverse exists because $u^\top B^\top H B u = (Bu)^\top H (Bu) > 0$ for nonzero u. Poorly scaled columns or a nearly singular metric still make it unstable. This projector is idempotent but generally not symmetric; $B^\top H(z - P_H z) = 0$.

The statistical orthogonality condition uses H. The cash exposure cancellation remains $B^\top w = 0$. Replacing it with $B^\top H w = 0$ without redefining the return coordinates changes the economic risk being neutralized.

Mathematical checkpoint

When the metric changes, which object should move: the statistical factor estimate, the return-loading map, or the portfolio constraint? Can all three be changed consistently?

4.4 Estimated subspaces move

A sample covariance changes when the training observations change. The economically relevant object is the retained subspace, so we compare projectors rather than attach meaning to an arbitrary eigenvector sign.

Explained variance is $\sum_{i \le k} \lambda_i / \operatorname{tr} \Sigma$, when $\operatorname{tr} \Sigma > 0$. A choice of k changes the residual dimension and the feasible neutral portfolio set. Choose it by a predeclared training criterion and forward prediction assessment, not by future realized gains.

Signs of eigenvectors are arbitrary; a repeated eigenvalue makes every orthonormal basis of its eigenspace equally valid. If two adjacent eigenvalues are nearly equal, individual directions are poorly determined even when their joint subspace is stable. For two orthonormal bases $V, \widetilde{V} \in \mathbb{R}^{n \times k}$, the singular values of $V^\top \widetilde{V}$ are $\cos \theta_i$, where $\theta_i \in [0, \pi/2]$ are principal angles. In equal dimension, $\|VV^\top - \widetilde{V}\widetilde{V}^\top\|_2 = \sin \theta_{\max}$. It is the projector that determines neutral exposures.

The eigengap $\delta = \lambda_k - \lambda_{k+1}$ controls sensitivity. Appendix B derives a first-order mixing coefficient proportional to $1/(\lambda_i - \lambda_j)$. Under a separated cluster and sufficiently small symmetric perturbation E, the subspace deviation is of order $\|E\|_2/\delta$. This is a local sensitivity statement, not a guarantee under structural breaks.

At a block boundary all estimates use only information already available. Within the block fix V_k, its residual complement $U \in \mathbb{R}^{n \times d}$, $d = n - k$, and the centering/scaling convention. Re-estimation starts a new coordinate system. A newly estimated factor is not substituted retroactively into a historical trading decision. A slow rotation can be studied through principal angles; a large rotation triggers the predeclared model review.

What we gained

- PCA derived from captured variance and reconstruction error.
- The SVD connection and the meaning of a weighted projection.
- A way to track subspace uncertainty and freeze trading coordinates.

Where this goes next

A small residual variance may look attractive, but it says nothing about a restoring force. We next construct a counterexample to the claim that PCA creates mean reversion.

Chapter 5

Why residuals need not mean revert

What problem are we solving?

We have removed common directions, not introduced temporal dependence. To avoid forecasting the wrong object, we distinguish a residual gain, its cumulative sum, and a combination of economic price levels.

5.1 A restoring force is an extra assumption

Let e_{t+1} be an independent centered residual gain with variance $s^2 > 0$. Even if s^2 is very small, the cumulative quantity $X_{t+1} = X_t + e_{t+1}$ wanders. Given the present, its expected next change is zero at every value of X_t. A large positive X_t does not make a fall more likely in this model.

Mean reversion would instead require, for example, $\mathbb{E}[X_{t+1} - X_t \mid X_t] = -\kappa X_t$, with $\kappa > 0$. The sign of this conditional change opposes the displacement. The equation is a statement about time, and must be justified by observations or an economic model. No identity for an orthogonal projection supplies it.

5.2 Three different residual objects

For the fixed residual basis U from Chapter 4, define the observed residual return $e_t = U^\top(r_t - \bar{r}) \in \mathbb{R}^d$. Its cumulative sum $X_t = X_0 + \sum_{s=1}^{t} e_s$ satisfies $\Delta X_t = e_t$. If e_t are independent centered innovations with covariance Q_e, then $\mathrm{Var}(X_t) = \mathrm{Var}(X_0) + tQ_e$, assuming X_0 is independent. Thus even perfectly white residual returns make a nonstationary cumulative residual.

Conversely, if $X_{t+1} = bX_t + \eta_{t+1}$ with $|b| < 1$, then $e_{t+1} = (b-1)X_t + \eta_{t+1}$ has a state-dependent mean. This is a substantive restriction on the integrated residual, not a result of PCA. It is falsified if the cumulative residual behaves like an unconfined random walk over the intended estimation horizon.

A third object is a cointegrating combination $\beta^\top p_t$, where $p_t \in \mathbb{R}^n$ denotes coherent economic price levels and $\beta \in \mathbb{R}^n$ is fixed over the research block. Log prices require positivity. A stationary combination of such levels is not generally a stationary combination of normalized settlement gains: changing multipliers, denominators and maturity selection break the equality. A cointegration strategy therefore requires a separately derived holdings map.

Common misconception

Small variance, zero contemporaneous factor exposure, stationarity and useful conditional prediction are four different properties. A model may have any one without the others.

What we gained

- A random-walk counterexample to automatic residual mean reversion.
- Distinct definitions of residual returns, accumulated residuals and level spreads.
- A conditional-mean equation that states a restoring-force hypothesis.

Where this goes next

A dynamic model must name the variable to which its hypothesis applies. We next decide what information to retain as the state and what stability means for that variable.

Chapter 6

Choosing an observable state

What problem are we solving?

Keeping the entire history makes forecasting unwieldy. We seek a small summary that retains the information our model needs, beginning with a state we can calculate directly from observations.

6.1 A state is a proposed summary

Suppose tomorrow's residual depends on today's residual alone. Then $x_t = e_t$ is a candidate state. If two recent residuals matter, use $x_t = (e_t, e_{t-1})^\top$. If an equilibrium spread matters, use its current level, provided its connection to actual gains has been derived. These are different hypotheses, not different notations for the same forecast.

The Markov property says that, conditional on the state, earlier observations do not further change the model's next-state probabilities. In a finite-state example it reads

$$\mathbb{P}(x_{t+1} = j \mid x_t = i, x_{t-1}, \ldots) = \mathbb{P}(x_{t+1} = j \mid x_t = i).$$

The numbers on the right form a transition matrix. In a continuous state they become probabilities of intervals or other sets. We will first derive an AR transition density; only later will we abbreviate such laws by a kernel.

6.2 What permits learning from one history?

We have one long sequence, not many independent copies of today's market. Stationarity asks whether shifting the time origin changes the relevant law. Ergodicity asks whether time averages along this one history recover the population quantities. Neither should be assumed just because a plot looks stable.

6.3 Stationarity and the right to average

Definition 6.1. A process x_t is strictly stationary if every finite joint law is invariant under a common time shift. It is weakly stationary if it has finite second moments, a constant mean, and covariance depending only on lag. An integrable stationary process is ergodic for an observable g if $N^{-1} \sum_{t=1}^{N} g(x_t) \to \mathbb{E}[g(x_0)]$ almost surely.

Strict stationarity with finite second moments implies weak stationarity. Weak stationarity alone does not guarantee a Gaussian stationary law or a law of large numbers. Estimating covariance from a single time path requires appropriate ergodicity of the relevant products. Confidence intervals additionally need a central limit theorem, often justified by stronger mixing and moment conditions.

A stationary mixture can fail to be ergodic: choose one unchanging latent regime at inception and remain there forever. Time averages converge to a regime-specific quantity rather than the unconditional mixture average. This matters because the researcher observes one realized history, not repeated universes.

6.4 A working choice, with a later refinement

For the first dynamic model take observed residual-return coordinates as the state. This gives a direct forecast of the next residual return. Later, noisy observations will motivate a hidden expected-return state. That extension will add a measurement equation only after we understand the observable model. Freeze the factor basis within an estimation/evaluation block so that a change in coordinate definitions cannot masquerade as a state innovation.

What we gained

- Observable states built from current or lagged residuals.
- The meaning of a Markov summary before kernel notation.
- Separate stationarity and ergodicity requirements for estimation.

Where this goes next

We have named a state but not specified how it moves. A scalar autoregression is the smallest model in which today influences tomorrow and that influence can fade.

Chapter 7

AR and VAR: forecasts by repeated substitution

What problem are we solving?

A state is useful only if it produces a conditional distribution. We start with one coefficient controlling persistence, work out its future effects, and then replace the coefficient by a matrix.

7.1 One-step persistence

Write $X_{t+1} = a + bX_t + \epsilon_{t+1}$. The intercept a sets the level, b transmits today's displacement, and ϵ_{t+1} is the new unpredictable innovation. Independence of that innovation from today's information makes its conditional mean zero. To see why powers of b appear, substitute once:

$$X_{t+2} = a(1 + b) + b^2 X_t + b\epsilon_{t+1} + \epsilon_{t+2}.$$

Old information is multiplied twice; the first new shock is multiplied once; the last shock has not yet decayed. The general horizon formula counts these same contributions. If X_t is a residual-return signal, its conditional mean is the candidate forecast that a portfolio may exploit.

7.2 Scalar autoregression

Consider $X_{t+1} = a + bX_t + \epsilon_{t+1}$, where ϵ_t are independent of the past, centered, with variance $\sigma_\epsilon^2 < \infty$. Repeated substitution gives

$$X_{t+h} = b^h X_t + a \sum_{j=0}^{h-1} b^j + \sum_{j=0}^{h-1} b^j \epsilon_{t+h-j}. \tag{7.1}$$

For $b \neq 1$, the geometric sum is $(1 - b^h)/(1 - b)$. If $|b| < 1$, let $\theta = a/(1 - b)$. Conditional mean and variance are

$$\mathbb{E}[X_{t+h} \mid X_t] = \theta + b^h(X_t - \theta), \qquad \operatorname{Var}(X_{t+h} \mid X_t) = \sigma_\epsilon^2 \frac{1 - b^{2h}}{1 - b^2}. \tag{7.2}$$

Under independent Gaussian innovations the transition is Gaussian. With other innovations it is the convolution law of the displayed weighted sum, not generally Gaussian.

The long-run interpretation

For $|b| < 1$, the conditional mean tends to θ, and conditional variance tends to $\sigma_\epsilon^2/(1-b^2)$. A stationary version has these moments. Its construction from an infinite past, convergence in mean square and non-Gaussian characteristic function are proved in Appendix C. Negative b gives alternating rather than monotone decay. These possibilities matter when translating a forecast into the sign and horizon of a position.

Estimating the transition from observed pairs

To estimate (a, b) from observations available through t, form rows $z_s^\top = (1, X_s)$ and minimize $\sum_s (X_{s+1} - z_s^\top \vartheta)^2$. The normal equations are $Z^\top Z\widehat{\vartheta} = Z^\top Y$, where Y stacks the responses. A unique estimate requires two linearly independent columns of Z. Nearly constant X_s makes the Gram matrix poorly conditioned. Gaussian likelihood gives the same coefficient estimate; non-Gaussian least squares estimates a conditional mean under zero conditional innovation mean. Stability must be constrained or tested, not manufactured by deleting an unfavorable estimate.

7.3 The scalar transition as a probability law

With Gaussian innovations of variance $\sigma_\epsilon^2 > 0$,

$$p(x' \mid x) = \frac{1}{\sqrt{2\pi\sigma_\epsilon^2}} \exp\left[-\frac{(x' - a - bx)^2}{2\sigma_\epsilon^2}\right].$$

Here x is the observed current state and x' a possible next state. Integrating over an interval gives its conditional probability. The later notation $K(x, A) = \int_A p(x' \mid x)\,\mathrm{d}x'$ is just this operation for a set A; K is called a transition kernel.

7.4 Vector autoregression

For two coordinates, the first row of A can use both current signals to predict the first future signal. Repeated multiplication by A therefore propagates both persistence and cross-coordinate effects. The innovation covariance Q records how simultaneous new shocks co-move; it is distinct from uncertainty about the estimated entries of A.

For a centered $x_t \in \mathbb{R}^\ell$, let $x_{t+1} = Ax_t + w_{t+1}$, with innovation covariance Q. Iteration gives

$$\mathbb{E}[x_{t+h} \mid x_t] = A^h x_t, \qquad Q_h = \mathrm{Cov}(x_{t+h} \mid x_t) = \sum_{j=0}^{h-1} A^j Q (A^j)^\top. \tag{7.3}$$

Derivation: the stable covariance

The spectral radius is the largest absolute eigenvalue. If the spectral radius $\rho(A) < 1$, a matrix norm bound $\|A^j\| \le C_\eta(\rho(A) + \eta)^j$, for sufficiently small $\eta > 0$, ensures convergence of the covariance series $P_\infty = \sum_{j\ge 0} A^j Q (A^j)^\top$. Shifting the index proves the discrete Lyapunov equation $P_\infty = AP_\infty A^\top + Q$. Uniqueness follows because the difference D of two solutions satisfies $D = A^j D (A^j)^\top \to 0$. With Gaussian innovations the invariant law is $\mathcal{N}(0, P_\infty)$; it may be degenerate if some directions receive no noise.

Stable eigenvalues do not prevent short-term amplification when A is nonnormal: $\|A^h\|_2$ can first grow and later decay. Cross-coordinate lead-lag effects favor a VAR over separate scalar models only if they improve conditional forecast performance after the larger parameter uncertainty is included.

For fully observed states, the least-squares transition estimate is

$$\widehat{A} = \left(\sum_s x_{s+1} x_s^\top\right)\left(\sum_s x_s x_s^\top\right)^{-1}. \tag{7.4}$$

The Gram matrix must be positive definite; otherwise identify only estimable directions or add a declared penalty. The residual outer products estimate Q, with degrees-of-freedom corrections depending on the fitted specification. Ergodicity and finite moments support consistency; dependent or heteroskedastic innovations change the sampling uncertainty.

What we gained

- One-step and multi-step AR laws, including non-Gaussian innovations.
- VAR forecast covariances and the discrete Lyapunov equation.
- Least-squares estimates with explicit rank and stability requirements.

Where this goes next

Decision intervals need not be equally spaced. A continuous-time model should explain how persistence and noise depend on elapsed time; the OU process supplies that extension.

Chapter 8

OU: mean reversion in continuous time

What problem are we solving?

The AR coefficient belongs to a chosen time step. We now want one law that supplies transitions over every elapsed time, and we need a precise way to accumulate continuously arriving noise.

8.1 From a small time step to a stochastic differential equation

Let $\kappa > 0$ measure the speed of pull toward a level θ, and let $\sigma \geq 0$ measure noise per square root of time. Over a small step Δ, consider

$$X_{t+\Delta} - X_t = \kappa(\theta - X_t)\Delta + \sigma\sqrt{\Delta}\, Z, \qquad Z \sim \mathcal{N}(0,1).$$

The drift is order Δ; noise is order $\sqrt{\Delta}$. That scaling lets variances add in proportion to elapsed time. Noise of order Δ would disappear when independent increments are refined indefinitely. The differential notation below is shorthand for an integral limit, not an ordinary derivative of a random path.

8.2 A concise stochastic-integral foundation

We construct the integral first for a position held constant over each interval. Choosing its value at the left endpoint expresses the same information rule as choosing a futures position before its next settlement.

A Brownian motion B_u is a continuous adapted process with $B_0 = 0$, independent Gaussian increments and $B_v - B_u \sim \mathcal{N}(0, v-u)$ for $v > u$. For a predictable step process $h_u = \sum_j h_j \mathbf{1}_{(u_j, u_{j+1}]}(u)$, with h_j known at u_j, define $\int h\,\mathrm{d}B = \sum_j h_j (B_{u_{j+1}} - B_{u_j})$. Independence and conditional centering remove cross terms, giving

$$\mathbb{E}\left[\left(\int_0^T h_u\,\mathrm{d}B_u\right)^2\right] = \mathbb{E}\int_0^T h_u^2\,\mathrm{d}u. \tag{8.1}$$

Approximation in this norm defines the integral for predictable square-integrable integrands. Its mean is zero. The notation $dX_u = b_u\,\mathrm{d}u + \sigma_u\,\mathrm{d}B_u$ means the corresponding integral equation. Standard existence and uniqueness hold for globally Lipschitz drift and diffusion with linear growth; the OU coefficients satisfy these conditions [17].

8.3 Scalar Ornstein–Uhlenbeck dynamics

The deterministic equation is solved by an integrating factor. The same method works here because the multiplying factor is a smooth deterministic function. The noise integral then has a variance determined by the isometry just established.

Let

$$\mathrm{d}X_u = \kappa(\theta - X_u)\,\mathrm{d}u + \sigma\,\mathrm{d}B_u, \qquad \kappa > 0, \quad \sigma \geq 0. \tag{8.2}$$

Derivation: the integrating factor

Multiplication by $e^{\kappa u}$, using the ordinary product rule with a differentiable deterministic factor, yields $\mathrm{d}[e^{\kappa u}(X_u - \theta)] = \sigma e^{\kappa u}\,\mathrm{d}B_u$. Integration from t to $t + h$ gives

$$X_{t+h} = \theta + e^{-\kappa h}(X_t - \theta) + \sigma \int_t^{t+h} e^{-\kappa(t+h-u)}\,\mathrm{d}B_u. \tag{8.3}$$

The future integral is independent of $\mathcal{F}_t$, centered Gaussian. Its variance, by (8.1), is

$$\sigma^2 \int_t^{t+h} e^{-2\kappa(t+h-u)}\,\mathrm{d}u = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa h}). \tag{8.4}$$

These two expressions derive the full conditional transition law. Letting $h \to \infty$ gives the invariant Gaussian law $\mathcal{N}(\theta, \sigma^2/(2\kappa))$. At $\kappa = 0$ the limit is Brownian motion with variance $\sigma^2 h$, without a finite stationary law.

The half-life solves $e^{-\kappa h_{1/2}} = 1/2$, hence $h_{1/2} = (\log 2)/\kappa$. It describes decay of a conditional mean displacement. A realized path need not hit its mean by that date. Costs, forecast revision, risk constraints and opportunity costs determine the holding policy, so half-life is not an instruction to liquidate.

At regular sampling interval Δ, the exact discrete transition has $b = e^{-\kappa\Delta} \in (0, 1)$, $a = \theta(1 - b)$, and innovation variance $\sigma_\epsilon^2 = \sigma^2(1 - b^2)/(2\kappa)$. Thus a negative AR coefficient cannot arise from this scalar OU. At one fixed interval a Gaussian AR with $0 < b < 1$ and an OU are observationally equivalent after this parameter transformation. Irregular intervals, continuous-time restrictions, and cross-horizon tests distinguish their use.

8.4 Vector OU and cointegration

The matrix version propagates shocks across residual directions. Error correction is a separate use of the same stable-recursion mathematics: its state is a combination of economic levels, so its holdings map must be rederived.

For $x_u \in \mathbb{R}^\ell$, let

$$\mathrm{d}x_u = K(\theta - x_u)\,\mathrm{d}u + L\,\mathrm{d}B_u, \qquad L \in \mathbb{R}^{\ell \times p}, \tag{8.5}$$

where B is a p-dimensional Brownian motion. Matrix multiplication by e^{Ku} gives conditional mean $\theta + e^{-Kh}(x_t - \theta)$ and covariance

$$Q_h = \int_0^h e^{-Ku} L L^\top e^{-K^\top u}\,\mathrm{d}u. \tag{8.6}$$

If every eigenvalue of K has strictly positive real part, the integral converges to P_∞. Integrating the derivative of $e^{-Ku} L L^\top e^{-K^\top u}$ proves $K P_\infty + P_\infty K^\top = L L^\top$. Gaussianity follows from the deterministic-kernel Brownian integral. Full support requires the covariance to be positive definite, which can hold through propagation even if L has fewer columns than rows.

A different state: error-correcting levels

For cointegration, suppose a level vector p_t is integrated of order one: its increments are stationary while its levels are not. A vector error-correction specification is

$$\Delta p_{t+1} = \alpha\beta^\top p_t + \sum_{j=1}^{p-1} \Gamma_j \Delta p_{t+1-j} + \eta_{t+1}, \tag{8.7}$$

with $\alpha, \beta \in \mathbb{R}^{n\times r_c}$, full column rank $r_c < n$, and $\Gamma_j \in \mathbb{R}^{n\times n}$. The integer p here is the lag order. For one lag, $z_t = \beta^\top p_t$ follows $z_{t+1} = (I_{r_c} + \beta^\top\alpha)z_t + \beta^\top\eta_{t+1}$. If this transition is stable, its conditional moments and stationary law follow from (7.3). The unspanned levels retain common stochastic trends. With additional lags, stack z_t and lagged increments into a companion state and require its stable roots after removing the common unit-root directions.

The conditional mean of Δp_{t+1} is the right side of (8.7) without η, and its conditional covariance is $\mathrm{Cov}(\eta)$ under homoskedasticity. Longer horizons follow by iterating the companion transition. Cointegration rank and vector estimates require their own nonstandard inference [6, 10]; treating estimated β as fixed understates uncertainty. The hypothesis is useful when economically comparable levels admit stable equilibrium combinations, not simply because PCA residual returns are small.

What we gained

- Brownian integration as a mean-square limit of nonanticipating sums.
- The exact scalar OU transition and its relation to an AR coefficient.
- Vector OU covariances and the separate error-correction construction.

Where this goes next

Both AR and OU so far treat the current state as observed. Market residuals may instead be noisy measurements of a persistent signal, which requires a distribution over the current state.

Chapter 9

Hidden states and noisy observations

What problem are we solving?

A large residual observation can mean a large underlying signal or simply a large measurement shock. Treating the observation as the signal would force the trading rule to react equally to both.

9.1 One unknown signal and one noisy reading

Let x be an unobserved signal and observe $y = x + v$, where v is centered measurement noise. The state x and the reading y are different random variables. Even after seeing y, uncertainty about x remains. For a finite illustration suppose x is either x_L or x_H, with prior probability p on x_H. Let $\ell_H(y)$ and $\ell_L(y)$ be the likelihoods of the observed reading under the two states. Bayes' rule gives

$$\mathbb{P}(x = x_H \mid y) = \frac{p\ell_H(y)}{p\ell_H(y) + (1-p)\ell_L(y)}.$$

The numerator weights the prior possibility by how well it explains the reading. The denominator normalizes the two weights to sum to one. This conditional distribution is a *belief*; it is not another unobserved market variable.

9.2 Filtering is repeated prediction and conditioning

If the signal changes through time, first propagate yesterday's belief using the state dynamics. Then condition on today's reading. Filtering means carrying out this update as observations arrive. It differs from smoothing, which uses later observations to revise a past state; smoothing cannot supply yesterday's real-time trading signal.

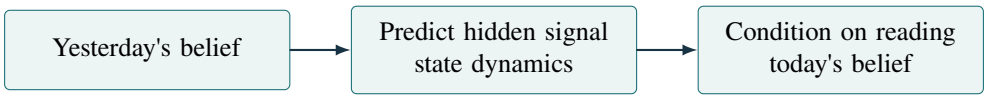


Figure 9.1: Only the state transition creates a new signal; the observation changes our information about it.

9.3 The maintained state for the main system

We now add the measurement equation to the already familiar linear recurrence. The matrix version allows fewer persistent hidden signals than observed residual-return coordinates.

The main model predicts residual *return* coordinates from a stable latent state:

$$x_{t+1} = Ax_t + w_{t+1}, \qquad e_t = Cx_t + v_t, \tag{9.1}$$

with $x_t \in \mathbb{R}^\ell$, $A \in \mathbb{R}^{\ell\times\ell}$, $C \in \mathbb{R}^{d\times\ell}$. The hidden state represents slowly evolving relative expected gains or equilibrium-pressure components observed through noisy residual returns. Its innovations w_t and measurement errors v_t have zero means and covariances $Q \in \mathbb{S}^\ell_+$, $R \in \mathbb{S}^d_{++}$. The Gaussian version assumes temporal independence and independence between the two sequences and the initial state.

Taking $C = I_d$ and observation noise zero yields an observable residual VAR as a special case, treated directly without inverting a zero observation covariance. Taking $e_{t+1} = C(x_{t+1} - x_t) + v_{t+1}$ instead makes the latent state a cumulative equilibrium deviation. That is a different observation equation, requiring different forecasts and an empirical comparison.

If the basis changes, define the new state explicitly. For cumulative levels p_t,

$$U_t^\top p_t - U_{t-1}^\top p_{t-1} = U_{t-1}^\top \Delta p_t + (U_t - U_{t-1})^\top p_t. \tag{9.2}$$

The second term is a coordinate-change term, not the gain of yesterday's hedge. Freezing the basis within a block avoids confusing this term with residual alpha.

Mathematical checkpoint

What observation would distinguish a slowly varying expected residual return from a stationary cumulative mispricing? Would their optimal positions have the same sign after a large residual move?

What we gained

- A distinction between a latent signal and its noisy observation.
- A belief obtained by ordinary Bayes weighting.
- A linear state and observation model with separate process and measurement noise.

Where this goes next

The two-state calculation updates probabilities explicitly. Gaussian assumptions give an equally explicit update for a continuous state: first a scalar weighted average, then the Kalman matrix recursion.

Chapter 10

Kalman filtering, one update at a time

What problem are we solving?

We need to combine a prediction and a noisy reading without either ignoring the observation or treating it as exact. Their relative uncertainties determine how much the estimate should move.

10.1 A scalar posterior by completing a square

Before observing y, suppose $x \sim \mathcal{N}(m^-, p^-)$, with $p^- > 0$. Let $y = x + v$, with independent $v \sim \mathcal{N}(0, r)$, $r > 0$. The superscript minus means "before this observation." Multiplication of the prior density by the observation likelihood gives an exponent proportional to

$$-\frac{1}{2}\left[\frac{(x-m^-)^2}{p^-} + \frac{(y-x)^2}{r}\right].$$

Derivation: precision determines the weights

Collect the coefficient of x^2, namely $1/p^- + 1/r$, and the coefficient of $-2x$, namely $m^-/p^- + y/r$. Completing the square therefore gives

$$p = \left(\frac{1}{p^-} + \frac{1}{r}\right)^{-1}, \qquad m = p\left(\frac{m^-}{p^-} + \frac{y}{r}\right) = m^- + \frac{p^-}{p^- + r}(y - m^-).$$

The new variance p is smaller than either input variance. The new mean is the old mean plus a fraction of the surprise $y - m^-$. Small measurement noise makes the fraction close to one; a precise prior makes it close to zero.

10.2 Prediction supplies the next prior

For $x_{t+1} = ax_t + w_{t+1}$, independent innovation variance q gives $m^-_{t+1} = am_t$ and $p^-_{t+1} = a^2 p_t + q$. Thus each cycle has two operations: dynamics can increase uncertainty, and an informative reading can decrease it. The trader uses the posterior mean to estimate the signal and the posterior variance to avoid treating that estimate as certain.

10.3 Kalman filtering from a joint Gaussian law

In several dimensions the scalar fraction becomes a matrix because each observation can inform several state directions. We derive that matrix by minimizing the error covariance, then use Gaussianity to identify it with the exact conditional distribution.

Use (9.1), now writing $y_t = e_t$, and initially suppose parameters are known. Let $m_{t-1} = \mathbb{E}[x_{t-1} \mid \mathcal{G}_{t-1}]$ and $P_{t-1} = \mathrm{Cov}(x_{t-1} \mid \mathcal{G}_{t-1})$. The prediction step follows by taking moments:

$$m_t^- = Am_{t-1}, \qquad P_t^- = AP_{t-1}A^\top + Q. \tag{10.1}$$

The new observation satisfies $y_t = Cx_t + v_t$. Define innovation $\zeta_t = y_t - Cm_t^-$ and innovation covariance $S_t = CP_t^-C^\top + R$. Conditional on $\mathcal{G}_{t-1}$, the joint vector (x_t, y_t) is Gaussian with mean (m_t^-, Cm_t^-) and block covariance

$$\begin{pmatrix} P_t^- & P_t^-C^\top \\ CP_t^- & S_t \end{pmatrix}. \tag{10.2}$$

Derivation: the best affine update

For any matrix $L_t \in \mathbb{R}^{\ell \times d}$, the error of an affine update $m_t^- + L_t\zeta_t$ has covariance

$$P_t^- - L_tCP_t^- - P_t^-C^\top L_t^\top + L_tS_tL_t^\top. \tag{10.3}$$

Completing the matrix square with $K_t = P_t^-C^\top S_t^{-1}$ gives

$$P_t^- - P_t^-C^\top S_t^{-1}CP_t^- + (L_t - K_t)S_t(L_t - K_t)^\top. \tag{10.4}$$

Since $R \succ 0$, $S_t \succ 0$, so the covariance-minimizing affine update is

$$m_t = m_t^- + K_t\zeta_t, \qquad P_t = P_t^- - P_t^-C^\top S_t^{-1}CP_t^-. \tag{10.5}$$

The residual $x_t - m_t^- - K_t\zeta_t$ is uncorrelated with ζ_t. Joint Gaussianity makes it independent of ζ_t, proving that (10.5) is the full posterior mean and covariance, not merely the best affine estimator. This is the probabilistic core of Kalman [12].

The equivalent covariance expression

$$P_t = (I - K_tC)P_t^-(I - K_tC)^\top + K_tRK_t^\top \tag{10.6}$$

exhibits positive semidefiniteness directly. Near-singular S_t means some observed combinations have almost no modeled uncertainty; inverse sensitivity then demands rechecking noise assumptions. With singular observation noise one must condition on its support or reduce redundant observations, rather than assert an inverse exists.

The prior is $\mathcal{N}(m_t^-, P_t^-)$, the likelihood is $y_t \mid x_t \sim \mathcal{N}(Cx_t, R)$, and the posterior is $\mathcal{N}(m_t, P_t)$. The gain K_t converts a surprise in observation units into a revision in latent-state units. A large state variance relative to observation noise makes the observation more influential, but the matrix direction matters.

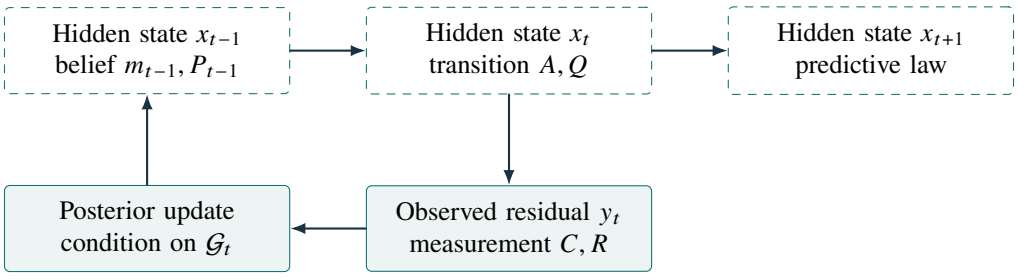


Figure 10.1: Latent state and observed residual. The lower return arrow denotes belief revision, not causation from the observation to the past market state.

What we gained

- A scalar Gaussian update derived from its density.
- Prediction and conditioning as distinct uncertainty operations.
- The matrix Kalman gain, posterior covariance and scope of exactness.

Where this goes next

The filter has treated its coefficients as known. A short history cannot identify them perfectly; the next chapter keeps uncertainty about parameters separate from uncertainty about the current state.

Chapter 11

Unknown parameters and competing models

What problem are we solving?

Knowing how to filter for a given persistence coefficient does not tell us which coefficient to use. We first compare two possible coefficients, then carry a distribution over parameters, and only then allow the model itself to change.

11.1 A coefficient is not a state innovation

Suppose an observed AR model has two possible persistence coefficients b_1, b_2, with prior weights ω_1, ω_2. If the observed transition is x to x', its density under choice j is $p_{b_j}(x' \mid x)$. The posterior weight is

$$\omega'_j = \frac{\omega_j p_{b_j}(x' \mid x)}{\omega_1 p_{b_1}(x' \mid x) + \omega_2 p_{b_2}(x' \mid x)}.$$

This is the same Bayes calculation used for hidden states. The difference is that an unknown fixed coefficient persists across all transitions. It is not drawn afresh whenever a new shock arrives. A future path must use the same coefficient until the model explicitly allows it to change.

11.2 What is a likelihood?

Regard the transition density as a function of the unknown coefficient with the observed data held fixed. That function is the likelihood. It measures how well a parameter explains those data; by itself it is not a probability density over parameters. Multiplying by a prior and normalizing produces that density.

11.3 Likelihood, restrictions and posterior uncertainty

Let $\vartheta \in \Theta$ collect the relevant coefficients and innovation parameters, with Θ enforcing positivity and the chosen stability restrictions. For observed transitions the likelihood through date t is

$$\mathcal{L}_t(\vartheta) = p_\vartheta(x_0) \prod_{s=0}^{t-1} p_\vartheta(x_{s+1} \mid x_s). \tag{11.1}$$

The symbol p_ϑ denotes a density relative to a declared dominating measure. A prior probability Π_0 produces $\Pi_t(\mathrm{d}\vartheta) = \mathcal{L}_t(\vartheta)\Pi_0(\mathrm{d}\vartheta)/\int_\Theta \mathcal{L}_t(\vartheta')\Pi_0(\mathrm{d}\vartheta')$ when the denominator is positive and finite. The symbol ϑ' is a dummy integration variable in the parameter space.

Maximum likelihood maximizes $\mathcal{L}_t$; a posterior averages plausible parameters. Both must account for the estimation of factors and the choice of state, either by conditioning honestly on a sealed training stage or by including those objects in the joint uncertainty. A stable fitted transition with highly uncertain coefficients is not strong evidence of a stable economic relation.

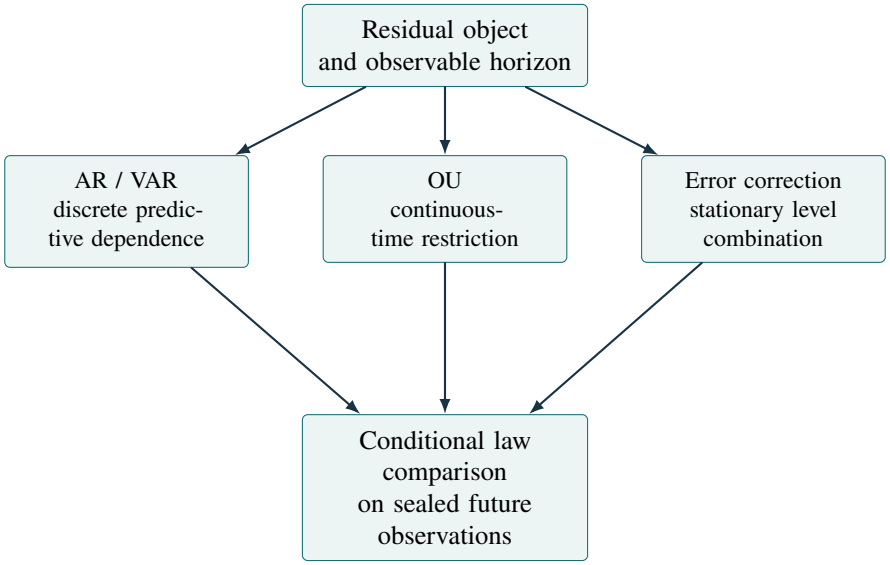


Figure 11.1: Candidate dynamics answer different empirical questions. A model is selected by the object it restricts and the forecasts it must defend.

11.4 Unknown parameters and general filtering

For a latent signal we must integrate over both the possible current signal and the possible parameters. In the notation below, K_ϑ is the transition law already illustrated by the AR density; ℓ_ϑ is the likelihood of a reading; b_t is their current joint posterior. The differential $\mathrm{d}x$ says to integrate over state possibilities, and $\mathrm{d}\vartheta$ says to integrate over parameter possibilities.

With unknown $\vartheta = (A, C, Q, R, \ldots)$, the innovation likelihood is $\prod_s \varphi_d(\zeta_s; 0, S_s)$, where φ_d is the d-dimensional Gaussian density and all filtered quantities depend on ϑ. This likelihood estimates structural parameters using only past observations. Plugging in a point estimate and reporting P_t alone omits parameter uncertainty.

For a general transition kernel $K_\vartheta(x, \mathrm{d}x')$ and observation density $\ell_\vartheta(y \mid x')$, define the joint belief $b_t(\mathrm{d}x, \mathrm{d}\vartheta) = \mathbb{P}(x_t \in \mathrm{d}x, \vartheta \in \mathrm{d}\vartheta \mid \mathcal{G}_t)$. Prediction and Bayes' rule give

$$b^-_{t+1}(\mathrm{d}x', \mathrm{d}\vartheta) = \int K_\vartheta(x, \mathrm{d}x') b_t(\mathrm{d}x, \mathrm{d}\vartheta), \tag{11.2}$$

$$b_{t+1}(\mathrm{d}x', \mathrm{d}\vartheta) = \frac{\ell_\vartheta(y_{t+1} \mid x') b^-_{t+1}(\mathrm{d}x', \mathrm{d}\vartheta)}{\int \ell_{\vartheta'}(y_{t+1} \mid z) b^-_{t+1}(\mathrm{d}z, \mathrm{d}\vartheta')}. \tag{11.3}$$

The denominator is the predictive observation density and must be positive and finite. Nonlinear or non-Gaussian transitions generally do not preserve a Gaussian posterior. The displayed integrals remain the mathematical problem; finite-dimensional Gaussian recursion is then an approximation whose forecast errors must be assessed.

11.5 A formal regime extension

Parameter uncertainty asks which fixed dynamics describe the block. A regime model asks whether the dynamics themselves switch. Begin with a finite label Z_t: predict its probabilities using a transition matrix, then update each weight using the observed transition likelihood. The generalization below is therefore another instance of the prediction–conditioning cycle.

Let $Z_t \in \{1, \dots, J\}$ be a latent Markov chain with transition matrix P^Z, where $P^Z_{ij} = \mathbb{P}(Z_{t+1} = j \mid Z_t = i)$. Assume nonnegative entries and row sums one. For the transparent observed-state extension, $x_{t+1} = A_{Z_{t+1}} x_t + \eta_{t+1}^{Z_{t+1}}$, with regime-$j$ Gaussian covariance $Q_j \in \mathbb{S}^\ell_{++}$, so its ordinary density exists, let $\alpha_{i,t} = \mathbb{P}(Z_t = i \mid \mathcal{G}_t)$. Then

$$\alpha^-_{j,t+1} = \sum_i \alpha_{i,t} P^Z_{ij}, \tag{11.4}$$

$$\alpha_{j,t+1} = \frac{\alpha^-_{j,t+1} \varphi_\ell(x_{t+1}; A_j x_t, Q_j)}{\sum_k \alpha^-_{k,t+1} \varphi_\ell(x_{t+1}; A_k x_t, Q_k)}. \tag{11.5}$$

This is the transition-plus-likelihood mechanism behind regime inference [9]. The next-state law is a mixture with mean $\bar{m} = \sum_j \alpha^-_j A_j x_t$ and covariance

$$\sum_j \alpha^-_j \left[Q_j + (A_j x_t - \bar{m})(A_j x_t - \bar{m})^\top \right]. \tag{11.6}$$

If the continuous state is also latent, the exact belief is a joint mixture over state and regime histories. Replacing it by one Gaussian per current regime discards history dependence and is an approximation.

Stability of every A_j alone does not guarantee stability under switching. A sufficient finite-second-moment condition is a common $H \succ 0$ and $c < 1$ such that $A_j^\top H A_j \preceq cH$ for every j, with uniformly bounded innovation second moments. Taking conditional expectations bounds the next squared H-norm by c times the current norm plus a constant. Together with irreducibility, aperiodicity and suitable nondegenerate noise, standard recurrence conditions yield an invariant law; it is generally not a Gaussian or a fixed mixture of the individual invariant Gaussians. The regime extension earns its complexity only through improved conditional calibration and decisions.

What we gained

- Likelihoods and posterior weights for unknown fixed parameters.
- The joint state–parameter filtering recursion.
- A regime extension with explicit mixture and stability qualifications.

Where this goes next

A belief about the present is not yet a forecast of tradable gains. We next propagate it into the future and reconstruct the original contract coordinates.

Chapter 12

The full predictive distribution

What problem are we solving?

The trader holds contracts, not latent coordinates. A useful forecast must return to the gain vector and must preserve uncertainty about the state, coefficients, common shocks and future sequence of observations.

12.1 Averaging conditional forecasts

For the two-parameter example, let m_j and s_j^2 be the next-gain mean and variance under choice j. The mixture mean is $\bar{m} = \sum_j \omega_j m_j$. Subtract and add m_j inside the squared error to obtain

$$\mathrm{Var}(Y) = \sum_j \omega_j s_j^2 + \sum_j \omega_j (m_j - \bar{m})^2.$$

The cross term vanishes because $\mathbb{E}[Y - m_j \mid j] = 0$. The first term is uncertainty within a model; the second is disagreement between forecasts. Reporting only a fitted model's variance discards the second contribution.

12.2 Recombining factor and residual forecasts

An orthonormal factor basis and its residual complement together span the original asset space. Recombination is therefore exact linear algebra; modeling the future coordinates is the uncertain part.

Complete V_k to an orthonormal basis (V_k, U). The exact coordinate identity in the frozen basis is

$$r_{t+1} = \bar{r} + V_k f_{t+1} + U e_{t+1}, \quad f_{t+1} = V_k^\top (r_{t+1} - \bar{r}), \quad e_{t+1} = U^\top (r_{t+1} - \bar{r}). \tag{12.1}$$

A joint model for (f_{t+1}, e_{t+1}) is needed even when the intended portfolio is factor neutral: loading error, imperfect execution and margin depend on common movement. For a base factor model use a stable VAR $f_{t+1} = A_f f_t + \xi_{t+1}$, estimated on past factor coordinates. Combine it with (9.1); allow a declared cross covariance between ξ_{t+1} and residual innovations if supported.

Let m_f, m_e be the conditional coordinate means, and let S_f, S_e, S_{fe} be the conditional covariance blocks given $\mathcal{G}_t$. Then

$$\mu_t = \bar{r} + V_k m_f + U m_e, \tag{12.2}$$

$$\Sigma_t = V_k S_f V_k^\top + U S_e U^\top + V_k S_{fe} U^\top + U S_{fe}^\top V_k^\top. \tag{12.3}$$

These follow from linearity and covariance expansion. They preserve positive semidefiniteness when the joint block covariance is positive semidefinite. Arbitrarily estimating the cross blocks independently may violate this requirement.

For known state parameters in the Gaussian main model,

$$m_e = CAm_t, \qquad S_e = C(AP_tA^\top + Q)C^\top + R. \tag{12.4}$$

The term involving P_t is latent-state uncertainty, Q is new process uncertainty, and R is observation variation. At horizon h, $\mathbb{E}[e_{t+h} \mid \mathcal{G}_t] = CA^h m_t$ and $\operatorname{Cov}(e_{t+h} \mid \mathcal{G}_t) = C(A^h P_t (A^h)^\top + Q_h)C^\top + R$. Cross-time covariance cannot be discarded when predicting a sequence of gains.

12.3 Path prediction and parameter uncertainty

For a multi-period position, we need a joint path law. Draw a current state and fixed parameter from the belief, propagate successive conditional transitions, then compute the holding gains along that path. The product below represents this ordered construction, with the intermediate states integrated out.

Let $\mathsf{P}_t = \mathcal{L}_{\mathbb{P}}(r_{t+1:t+h} \mid \mathcal{G}_t)$, a probability measure on $(\mathbb{R}^n)^h$. With state path $x_{t:t+h}$, parameters ϑ, and the joint return-state transition K_ϑ, the predictive law is

$$\mathsf{P}_t(\,\mathrm{d}r_{t+1:t+h}) = \int b_t(\,\mathrm{d}x_t,\,\mathrm{d}\vartheta) \prod_{j=1}^{h} K_\vartheta(x_{t+j-1},\,\mathrm{d}x_{t+j},\,\mathrm{d}r_{t+j}). \tag{12.5}$$

State coordinates in the product are integrated out. This notation permits dependence between next state and next gain. A product of unconditional one-step marginal densities would incorrectly erase serial dependence.

For any scalar future gain Y, let $m(\vartheta) = \mathbb{E}[Y \mid \mathcal{G}_t, \vartheta]$ and $v(\vartheta) = \operatorname{Var}(Y \mid \mathcal{G}_t, \vartheta)$. The tower property yields

$$\mathbb{E}[Y \mid \mathcal{G}_t] = \int m(\vartheta) b_t(\,\mathrm{d}\vartheta), \tag{12.6}$$

$$\operatorname{Var}(Y \mid \mathcal{G}_t) = \int v(\vartheta) b_t(\,\mathrm{d}\vartheta) + \operatorname{Var}_{b_t}(m(\vartheta)). \tag{12.7}$$

The second term is parameter-induced dispersion of forecasts. It vanishes under a plug-in estimate even when economically large. A Bayesian expected-utility decision integrates over this law; it need not randomize its action.

For a fixed exposure held over h periods with a fixed normalization, the covariance of cumulative gains is $\sum_{i,j=1}^{h} \operatorname{Cov}(r_{t+i}, r_{t+j} \mid \mathcal{G}_t)$. Multiplying one-step covariance by h is valid only if all cross-time covariances vanish and marginal covariances agree. If exposure, normalization or margin changes, compute the gain through the wealth recursion instead.

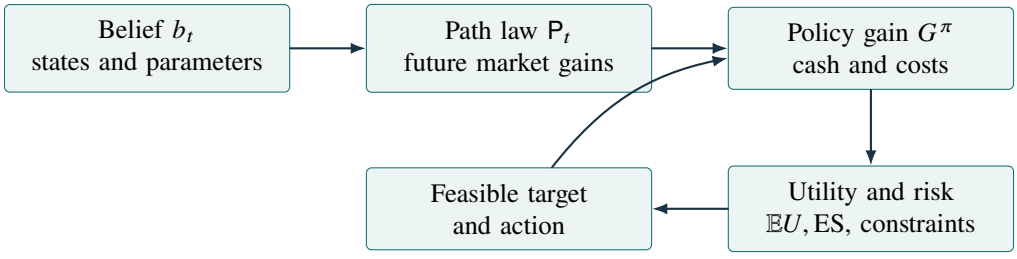


Figure 12.1: A forecast becomes a position only after a gain map, objective and feasible set have been specified.

What we gained

- A gain forecast in the original contract coordinates.
- A variance decomposition retaining state and parameter uncertainty.
- A joint predictive path law that preserves serial dependence.

Where this goes next

A predictive law contains more information than a covariance matrix. We now decide which summaries describe dispersion, extreme loss and funding risk, and what their estimation can support.

Chapter 13

Covariance, tails and path risk

What problem are we solving?

A forecast with a small mean-square error can still underestimate rare losses or the cash required before recovery. We separate covariance, tail loss and path risk instead of using one number for all three.

13.1 What a covariance says about a position

If the gain vector has conditional covariance Σ, then an exposure w has conditional gain variance $w^\top \Sigma w$. This follows by expanding $\mathbb{E}[(w^\top (r-\mu))^2 \mid \mathcal{G}_t]$. Off-diagonal entries matter: two uncertain gains may offset or reinforce each other. The eigenvalues measure uncertainty along orthogonal directions, not the reliability with which those directions have been estimated.

13.2 Why sample covariance can fail

Given N_{tr} observations in the training set, define

$$\widehat{\Sigma} = \frac{1}{N_{\mathrm{tr}} - 1} \sum_s (r_s - \widehat{m})(r_s - \widehat{m})^\top, \qquad \widehat{m} = \frac{1}{N_{\mathrm{tr}}} \sum_s r_s. \tag{13.1}$$

This is a descriptive covariance of the training distribution. Using it as tomorrow's conditional covariance requires a local homogeneity assumption or an explicit volatility-state adjustment. Its rank is at most $N_{\mathrm{tr}} - 1$. Thus it is singular when the asset dimension exceeds that number, regardless of how much algebra follows.

Even when invertible, small eigenvalues can be mostly estimation noise. In the mean-variance solution, an estimated mean component along eigenvector v_i is divided by λ_i. A tiny noisy eigenvalue therefore creates a large position. Temporal dependence reduces effective information, and heavy tails can make the fourth moments needed for covariance uncertainty infinite.

Choose a target $T \succ 0$, such as a diagonal matrix of positive variance estimates, and

$$\widehat{\Sigma}_\lambda = (1-\lambda)\widehat{\Sigma} + \lambda T, \qquad 0 < \lambda \le 1. \tag{13.2}$$

For any $u \neq 0$, $u^\top \widehat{\Sigma}_\lambda u \ge \lambda \lambda_{\min}(T)\|u\|^2 > 0$. The lower eigenvalue bound stabilizes the inverse. If $T = \tau I$, the quadratic risk penalty adds $\gamma\lambda\tau\|w\|^2/2$, revealing the regularization interpretation.

Derivation: the shrinkage criterion

The Frobenius norm satisfies $\|H\|_F^2 = \sum_{ij} H_{ij}^2$, with inner product $\langle H, J\rangle_F = \operatorname{tr}(H^\top J)$. It measures squared matrix-entry error. To see how a shrinkage weight can be principled, suppose temporarily T is fixed and Σ is the population covariance. Put $E = \widehat{\Sigma} - \Sigma$ and $D = T - \widehat{\Sigma}$. Expected squared Frobenius error is $\mathbb{E}\|E + \lambda D\|_F^2$. Differentiating gives the unconstrained oracle weight

$$\lambda_{\text{or}} = -\frac{\mathbb{E}\langle E, D\rangle_F}{\mathbb{E}\|D\|_F^2}, \tag{13.3}$$

projected onto $[0, 1]$, provided the denominator is positive. This quantity is unknown because it contains Σ; Ledoit and Wolf [13] develops estimable shrinkage under explicit asymptotic assumptions. With dependence or adaptive targets, estimate the criterion on past validation blocks or account for those changes in the moment calculation. A target is a bias assumption, not a cure for every distributional failure.

A factor risk estimator $B\widehat{\Sigma}_f B^\top + \widehat{D}_\varepsilon$ is positive definite if $\widehat{D}_\varepsilon \succ 0$. It reduces parameters by assuming a structure for residual dependence. Its empirical challenger is a shrinkage estimator that allows broader residual correlation.

13.3 Tails and dependence

Equal covariances do not imply equal probabilities of very large loss. A random common scale gives a simple way for otherwise mild shocks to become large together. We introduce this scalar mechanism before using its vector version.

A multivariate Gaussian $\mathcal{N}(\mu, \Sigma)$ is determined by its first two moments and has exponentially decaying squared tails. An elliptical random vector can be represented as $Y = \mu + RAu$, where u is uniform on the unit sphere, $R \geq 0$ is an independent radial variable and A is a deterministic matrix. Its covariance, when finite, is $\mathbb{E}[R^2]AA^\top/d$ in d dimensions. The scatter matrix $AA^\top$ need not equal covariance.

A Student law has the scale-mixture representation

$$Y = \mu + \frac{LZ}{\sqrt{\Lambda}}, \quad Z \sim \mathcal{N}(0, I_d), \quad \Lambda \sim \text{Gamma}(\nu/2, \nu/2), \quad Z \perp \Lambda. \tag{13.4}$$

The Gamma parameters are shape and rate; $LL^\top$ is a scale matrix. For $\nu > 2$, $\operatorname{Cov}(Y) = \nu LL^\top/(\nu-2)$, because $\mathbb{E}[\Lambda^{-1}] = \nu/(\nu-2)$. For $\nu \leq 2$ variance-based optimization has no population covariance under this model. A common random scale creates joint tail dependence even if $LL^\top$ is diagonal.

Finite mixtures $p(y) = \sum_j \omega_j p_j(y)$, with nonnegative weights summing to one, can capture asymmetry, multimodality and regime uncertainty. Their covariance contains both average component covariance and dispersion of component means. A compensated jump component, with intensity λ_J and jump-size law having finite second moment, adds rare discontinuities not described by diffusion variance alone. Its use requires observable evidence and a stress policy.

The linear conditional-mean recursions need centered innovations, and covariance recursions need second moments and stated independence. Exact Gaussian transition laws and exact Gaussian posterior filtering need Gaussianity. Tail probabilities, expected shortfall and exponential utility cannot generally be recovered from mean and covariance.

13.4 Why a quantile is not the whole tail

A loss quantile marks a threshold; it does not describe the severity of losses beyond that threshold. Expected shortfall averages the upper tail, including a fraction of any probability mass at the threshold

when necessary. The optimization formula below handles that fraction without a special case for each discrete loss distribution.

13.5 Variance, tails and path risk

For a loss $L = -G$, define $\mathrm{VaR}_\alpha(L) = \inf\{\ell : \mathbb{P}(L \le \ell \mid \mathcal{G}_t) \ge \alpha\}$, with $0 < \alpha < 1$. Define upper-tail expected shortfall by

$$\mathrm{ES}_\alpha(L) = \inf_{z \in \mathbb{R}} \left\{ z + \frac{1}{1-\alpha} \mathbb{E}[(L - z)_+ \mid \mathcal{G}_t] \right\}. \tag{13.5}$$

For an integrable loss this definition handles atoms correctly. If the law is continuous, differentiating the criterion gives $1 - \mathbb{P}(L > z)/(1-\alpha) = 0$, so a minimizing z is an α-quantile. At atoms use left and right derivatives. Convexity of positive part proves convexity in an affine portfolio loss; this is the decision advantage studied by Rockafellar and Uryasev [16].

For Gaussian portfolio loss with mean m_L and standard deviation $s_L > 0$, $\mathrm{ES}_\alpha = m_L + s_L \varphi(\Phi^{-1}(\alpha))/(1-\alpha)$, where Φ, φ are standard normal distribution and density. This follows by integrating $\int_z^\infty u\varphi(u)\,du = \varphi(z)$, since $\varphi'(u) = -u\varphi(u)$. Using this formula for a heavy-tailed or regime mixture loss replaces its tails by an unjustified Gaussian approximation.

Let $H_t^W = \max_{s \le t} W_s$ be a running wealth high and $D_t^W = H_t^W - W_t$ be absolute drawdown. Its update is $H_{t+1}^W = \max(H_t^W, W_{t+1})$. A drawdown constraint requires this path state, unlike a one-period variance bound. A chance constraint $\mathbb{P}(D_{t+1}^W > d_{\max} \mid \mathcal{G}_t) \le \epsilon$ is not equivalent to a hard bound, and need not be convex.

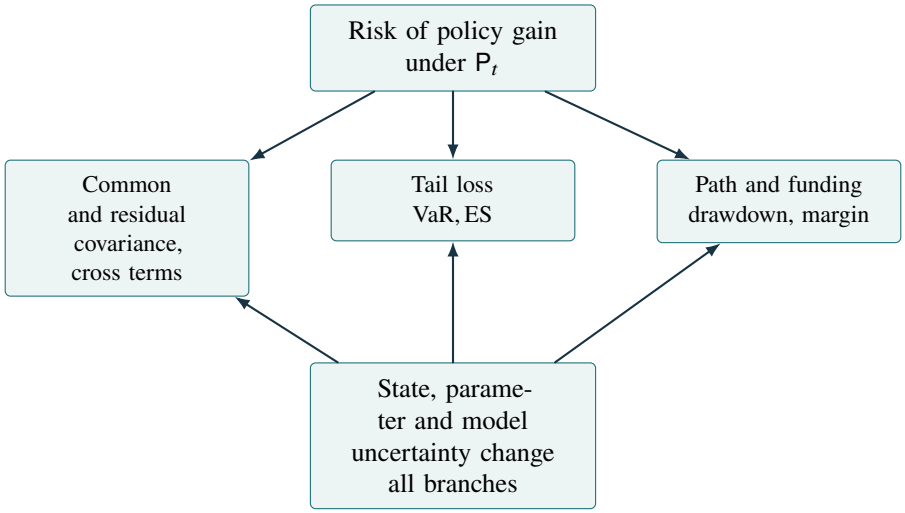


Figure 13.1: Risk has distinct distributional and path-dependent meanings. These are not interchangeable names for covariance.

What we gained

- Conditional covariance and its estimation/conditioning limits.
- Heavy-tailed and mixture alternatives to Gaussian shocks.
- Expected shortfall, drawdown and funding events as distinct risk objects.

Where this goes next

Risk summaries still do not choose a position. A preference must specify how much expected gain is worth how much uncertainty; we begin with a scalar mean–variance decision.

Chapter 14

Turning a forecast into a position

What problem are we solving?

A positive conditional mean gives no position size. Doubling a holding doubles expected gain but quadruples its variance. We make this tradeoff explicit and then connect the chosen exposure to contract counts.

14.1 One signal, one risk preference

Let a normalized gain r have conditional mean μ and variance $s^2 > 0$. For an exposure w, consider

$$J(w) = w\mu - \frac{\gamma}{2}w^2 s^2, \qquad \gamma > 0.$$

The first term rewards expected gain; the second penalizes dispersion. The constant γ is a preference expressed in these gain units. Differentiating gives $J'(w) = \mu - \gamma s^2 w$, so $w^* = \mu/(\gamma s^2)$. Since $J'' = -\gamma s^2 < 0$, this stationary point is the unique maximum. Increasing risk aversion or variance reduces the holding. A zero mean gives a zero holding in this unconstrained laboratory.

14.2 Why correlation changes the answer

With several assets, the derivative of the quadratic risk term is $\gamma\Sigma w$. Its i-th coordinate includes the risk contribution of every other holding correlated with asset i. Thus a position cannot generally be sized by dividing each mean separately by its marginal variance.

14.3 Units and the mean-variance laboratory

At date t, choose a strictly positive reference capital A_t, known in $\mathcal{G}_t$. Define normalized dollar exposure $w = D_t q_t^+ / A_t$ and incoming exposure $w^- = D_t q_t^- / A_t$. Then cash market gain divided by A_t is $w^\top r_{t+1}$. Reference capital is held fixed within this date's decision; it is not necessarily all current wealth. A negative equity state invokes the previously specified liquidation rule instead of redefining leverage with a negative denominator.

The mean-variance criterion is

$$J(w) = w^\top \mu - \frac{\gamma}{2}w^\top \Sigma w, \qquad \gamma > 0. \tag{14.1}$$

It is a preference specification. For Gaussian gain and exponential utility $U(g) = -e^{-\gamma g}$, direct Gaussian integration gives $\mathbb{E}[U(w^\top r)] = -\exp\{-\gamma w^\top \mu + \gamma^2 w^\top \Sigma w/2\}$. Maximizing this expectation is equivalent to (14.1). For non-Gaussian gains it is generally a second-order approximation or a separate moment preference, not exact expected utility.

If $\Sigma \succ 0$, stationarity gives $\mu - \gamma \Sigma w = 0$, and the negative definite Hessian establishes

$$w_0 = \gamma^{-1} \Sigma^{-1} \mu. \tag{14.2}$$

If Σ is singular and μ has a nonzero component in its nullspace, the unconstrained objective is unbounded above along that direction. If μ is orthogonal to the nullspace, maximizers may be nonunique. A covariance inverse therefore represents both a mathematical condition and an economic fragility.

Why this matters

The inverse covariance transforms a forecast into a risk-adjusted exposure. It does not improve the forecast. A noisy mean combined with a nearly singular covariance can produce a very precise-looking but unstable recommendation.

What we gained

- A scalar position derived from an explicit preference.
- The multivariate mean–variance solution and its utility interpretation.
- A consistent conversion between capital, normalized exposure and contracts.

Where this goes next

The unconstrained optimizer may use too much capital or take unwanted factor risk. We next restrict the available holdings and derive how those restrictions alter the optimum.

Chapter 15

Constraints and factor-neutral portfolios

What problem are we solving?

A desirable exposure can be unfinanceable or conflict with a hedge requirement. We use the three-asset neutral plane to understand constrained optimization before adding a full system of limits.

15.1 Optimizing inside the neutral plane

Return to the one-factor loading vector β, with $\beta_3 \neq 0$, and set

$$N = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -\beta_1/\beta_3 & -\beta_2/\beta_3 \end{pmatrix}.$$

Every neutral exposure is $w = Nz$, where $z \in \mathbb{R}^2$ contains the two free holdings. Substitution into the objective gives

$$J(Nz) = z^\top N^\top \mu - \frac{\gamma}{2} z^\top N^\top \Sigma N z.$$

If the reduced covariance $N^\top \Sigma N$ is positive definite, ordinary differentiation yields $z^* = \gamma^{-1}(N^\top \Sigma N)^{-1} N^\top \mu$. This matrix describes uncertainty in the bets that remain after hedging. Nothing requires N to be orthonormal; using an orthonormal basis changes coordinates, not the feasible plane.

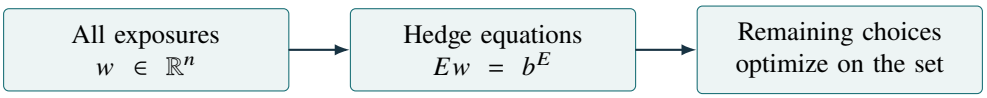


Figure 15.1: Constraints remove degrees of freedom; the forecast and objective select among those still available.

15.2 Equality constraints in closed form

For an arbitrary affine hedge target, direct substitution can be cumbersome. Lagrange multipliers derive the same risk-metric projection without choosing a nullspace basis.

With only $Ew = b^E$, assume $\Sigma \succ 0$ and E has full row rank. Stationarity gives $w = w_0 - \gamma^{-1}\Sigma^{-1}E^\top \nu$. Enforcing the equality yields $\nu = \gamma(E\Sigma^{-1}E^\top)^{-1}(Ew_0 - b^E)$, hence

$$w^\star = w_0 - \Sigma^{-1}E^\top (E\Sigma^{-1}E^\top)^{-1}(Ew_0 - b^E). \tag{15.1}$$

For $z \neq 0$, $z^\top E\Sigma^{-1}E^\top z > 0$ because $E^\top z \neq 0$, proving the second inverse exists. Nearly redundant equality rows still cause poor conditioning. Formula (15.1) is the projection of w_0 onto the affine feasible set under the Σ-metric, not generally the Euclidean projection.

Completing the square gives $J(w) = J(w_0) - \frac{\gamma}{2}(w - w_0)^\top \Sigma (w - w_0)$. This explains why equality constraints remove the least valuable directions according to risk geometry. Extra active inequalities require solving their KKT system, with no general closed form.

15.3 A complete one-period feasible set

Build the following set one row at a time: equality targets, leverage, individual and group capacity, risk, and finally current cash. Here W_t is current equity, K_t a cash trading cost, and $\mathfrak{m}_t$ required collateral. Each restriction answers a different operational question.

Let $E_t \in \mathbb{R}^{p_E \times n}$ contain linearly independent equality rows, including chosen factor-neutral rows $B_t^\top$, and let $b_t^E \in \mathbb{R}^{p_E}$ be their target. This b_t^E is not the filtering belief b_t. Let $S_{\text{sec}} \in \mathbb{R}^{p_S \times n}$ map exposures to asset groups. Define

$$\begin{aligned} C_t = \{w \in \mathbb{R}^n : {} & E_t w = b_t^E, \quad \|w\|_1 \le L_t, \quad \ell_t^{\text{net}} \le \mathbf{1}^\top w \le u_t^{\text{net}}, \\ & |w_i| \le \bar{w}_{i,t}, \quad \ell_t^{\text{sec}} \le S_{\text{sec}} w \le u_t^{\text{sec}}, \\ & w^\top \widehat{\Sigma}_t w \le \bar{\sigma}_t^2, \quad \|D_{\text{con}} w\|_2 \le c_t, \\ & \widehat{\mathfrak{m}}_t(A_t D_t^{-1} w) + K_t(A_t D_t^{-1}(w - w^-)) \le W_t - C_t^{\text{reserve}}\}. \end{aligned} \tag{15.2}$$

All bounds are $\mathcal{G}_t$-measurable and fixed before optimization. D_{con} is a diagonal concentration weighting matrix; $\bar{\sigma}_t$ is a risk cap, not an equality forcing the system to take risk. The vector $\mathbf{1}$ consists of ones. The margin map $\widehat{\mathfrak{m}}_t$ must be a declared convex conservative envelope of actual requirements for convex analysis to apply. A scenario maximum plus nonnegative absolute position charges is one such model; exchange margin need not itself have this form.

Include cost and turnover in the objective:

$$\max_{w \in C_t} \left\{ \mu_t^\top w - \frac{\gamma}{2} w^\top \widehat{\Sigma}_t w - c_t^{\text{tr}}(w - w^-) - \eta_t \|w - w^-\|_1 \right\}. \tag{15.3}$$

Here $c_t^{\text{tr}} = R_t^f K_t / A_t$ is a cost in the same normalized end-period gain units and $\eta_t \ge 0$ is an additional turnover preference, not another estimate of the same cost. One useful convex cost is $c_t^{\text{tr}}(u) = s_t^\top |u| + \frac{1}{2} u^\top \Lambda_t u$, with $s_t \ge 0$ and $\Lambda_t \succeq 0$. If costs are already realized inside the return gain map, they are not subtracted here again.

Every added restriction has a separate role: gross exposure limits leverage, net exposure controls signed imbalance, factor rows remove designated common movement, group bounds restrict economic concentration, individual bounds restrict capacity, the covariance cap limits modeled dispersion, and the margin inequality preserves present funding feasibility. None implies the others.

If C_t is nonempty, closed and bounded and costs are continuous, a maximum exists. With $\widehat{\Sigma}_t \succ 0$, the objective is strictly concave on a convex feasible set and the continuous optimum is unique. If the constraints conflict, there is no optimizer; the policy must reduce optional targets or liquidate according to a declared priority, never silently report an infeasible position.

15.4 KKT conditions and shadow prices

When an inequality binds, a small relaxation may improve the objective. Its multiplier measures that local value. Complementary slackness says a loose constraint has no such marginal price under the stated regularity assumptions.

Write the minimization version as $f(w) = \frac{\gamma}{2} w^\top \Sigma w - \mu^\top w + c(w - w^-)$, with convex inequalities $g_j(w) \le 0$ and equality $Ew = b^E$. The Lagrangian is $\mathcal{L}(w, \lambda, \nu) = f(w) + \sum_j \lambda_j g_j(w) + \nu^\top (Ew - b^E)$. For differentiable g_j, necessary and sufficient optimality conditions under convexity and a relative-interior Slater point are

$$0 \in \gamma \Sigma w - \mu + \partial c(w - w^-) + \sum_j \lambda_j \nabla g_j(w) + E^\top \nu, \tag{15.4}$$

$$g_j(w) \le 0, \quad Ew = b^E, \quad \lambda_j \ge 0, \quad \lambda_j g_j(w) = 0. \tag{15.5}$$

For a weighted absolute cost, its i-th subgradient equals $s_i \operatorname{sign}(u_i)$ when $u_i \neq 0$, and lies in $[-s_i, s_i]$ at zero. This interval creates a no-trade region.

Weak duality follows since $\lambda_j g_j(w) \le 0$ at any feasible w. Taking the infimum of $\mathcal{L}$ over w gives a lower bound on the minimum. Slater's condition gives equality of primal and dual values. If a bound $g_j(w) \le b_j$ is relaxed, the optimal minimized cost changes locally at rate $-\lambda_j$, where differentiability holds. The multiplier is the marginal value of that extra risk or capital capacity, conditional on the model.

15.5 Contract discreteness

Actual contracts require $q \in \mathbb{Z}^n$, so the exposure lattice is $\{D_t q / A_t : q \in \mathbb{Z}^n\}$. The final contract decision is the maximizer of the same criterion over this lattice intersected with feasible constraints. Bounded contract limits make this a finite set, so an optimum exists if the set is nonempty; use a fixed tie-break rule.

Rounding a continuous optimum can violate neutrality and margin. Exact factor equalities with irrational estimated loadings can leave only the zero integer portfolio. Operational neutrality is therefore specified as a tolerance $\|B_t^\top D_t q / A_t\|_\infty \le \epsilon_B$, unless exact lattice feasibility has been established. The tolerance is a research parameter, included in validation and risk, not an invisible rounding error.

What we gained

- Nullspace coordinates for the bets left after hedging.
- An equality-constrained optimum and KKT conditions for inequalities.
- A feasible contract problem that respects margin and integer quantities.

Where this goes next

Feasibility does not make estimated inputs correct. We next ask how the position changes when the forecast itself is allowed to vary within a declared uncertainty set.

Chapter 16

Robust optimization and the decision not to trade

What problem are we solving?

An optimizer can exploit estimation error as aggressively as a real signal. We describe uncertainty in the mean explicitly and derive the penalty that protects against its unfavorable directions.

16.1 The scalar uncertainty interval

Suppose the mean lies in $[\widehat{\mu} - \rho, \widehat{\mu} + \rho]$. For $w > 0$, the lowest mean hurts most; for $w < 0$, the highest mean does. Both cases are summarized by

$$\inf_{|\mu - \widehat{\mu}| \le \rho} w\mu = w\widehat{\mu} - \rho|w|.$$

With cost $s|w|$ from a zero position and variance penalty $\gamma\sigma^2 w^2/2$, the best holding is zero whenever $|\widehat{\mu}| \le \rho + s$. A small estimated edge can disappear completely once uncertainty and the cost of reaching it are acknowledged.

16.2 Mean ambiguity and support functions

An ellipsoid generalizes the interval: its shape describes which mean directions are uncertain, and its support function computes the worst exposure-weighted error. Cauchy–Schwarz supplies the calculation.

Let the mean belong to $\mathcal{U}_\mu = \{\widehat{\mu} + Lu : \|u\|_2 \le \rho_\mu\}$, where $L \in \mathbb{R}^{n \times p}$ describes estimation-error directions and $\rho_\mu \ge 0$ their radius. Cauchy–Schwarz gives

$$\inf_{\mu \in \mathcal{U}_\mu} w^\top \mu = w^\top \widehat{\mu} + \inf_{\|u\| \le \rho_\mu} (L^\top w)^\top u \tag{16.1}$$

$$= w^\top \widehat{\mu} - \rho_\mu \|L^\top w\|_2. \tag{16.2}$$

When $L^\top w \neq 0$, equality is attained at $u = -\rho_\mu L^\top w / \|L^\top w\|$; if it is zero, every feasible u gives the same mean contribution. Thus a norm penalty expresses a precise worst-case mean belief [8].

For covariance uncertainty $\mathcal{U}_\Sigma = \{\widehat{\Sigma} + E : E = E^\top,\ \|E\|_2 \le \rho_\Sigma,\ \widehat{\Sigma} + E \succeq 0\}$,

$$\sup_{\Sigma \in \mathcal{U}_\Sigma} w^\top \Sigma w = w^\top \widehat{\Sigma} w + \rho_\Sigma \|w\|_2^2. \tag{16.3}$$

The upper bound is the operator-norm inequality and is attained by the positive semidefinite perturbation $E = \rho_\Sigma w w^\top / \|w\|^2$ for $w \neq 0$. The robust criterion becomes

$$\widehat{\mu}^\top w - \rho_\mu \|L^\top w\|_2 - \frac{\gamma}{2} w^\top (\widehat{\Sigma} + \rho_\Sigma I) w - c(w - w^-). \tag{16.4}$$

This derivation explains one robustness interpretation of regularization. It does not imply that every penalty corresponds to a correctly calibrated confidence set.

16.3 Sensitivity and the no-position decision

For the unconstrained optimum, a small perturbation satisfies to first order

$$\delta w = \gamma^{-1} \Sigma^{-1} \delta\mu - \Sigma^{-1} (\delta\Sigma) w. \tag{16.5}$$

To derive it, differentiate $\gamma \Sigma w = \mu$ and solve the resulting linear equation. Large $\|\Sigma^{-1}\|$, large positions, or nearly redundant constraints amplify uncertainty. Stability of a recommendation is therefore part of its economic interpretation.

With zero incoming position, homogeneous costs $s^\top |w|$, zero equality targets, and a feasible zero portfolio, there is no beneficial infinitesimal trade if

$$\widehat{\mu}^\top d \leq \rho_\mu \|L^\top d\| + s^\top |d| \quad \text{for every feasible direction } d. \tag{16.6}$$

Because the quadratic risk penalty is nonnegative, this condition also makes zero globally optimal for the homogeneous feasible setting. Uncertainty can remove a trade entirely, not merely shrink it.

A Bayesian rule maximizes expected utility under the posterior predictive law. A robust rule minimizes over an ambiguity set. They answer different questions: posterior probabilities weigh alternatives, while robustness protects against an allowed adversarial alternative. Using both is meaningful only when the posterior and the remaining misspecification set have distinct roles.

Mathematical checkpoint

If the entire apparent edge lies along the least accurately estimated covariance direction, what evidence would justify increasing the position? How does that question change when costs are linear near zero?

What we gained

- A scalar no-position threshold combining costs and estimation uncertainty.
- Norm and quadratic penalties derived from explicit ambiguity sets.
- A distinction between posterior averaging and worst-case protection.

Where this goes next

All these decisions have treated the next interval in isolation. Trading today also changes the inventory and cash available tomorrow; that link requires a policy across time.

Chapter 17

Multi-period decisions and the Bellman principle

What problem are we solving?

A trade that looks attractive for one interval may be costly to reverse. We need to value its immediate gain together with the choices it leaves available after the next observation.

17.1 A two-period trading problem

For two intervals, let a_0 be the first trade and a_1 the trade chosen after seeing the next state s_1. That state includes both the signal and the inventory inherited from a_0. Let R_0, R_1 be net stage rewards and $\Phi(s_2)$ a terminal value. Backward reasoning starts with

$$V_1(s_1) = \max_{a_1 \in \mathcal{A}_1(s_1)} \mathbb{E}[R_1 + \Phi(s_2) \mid s_1, a_1].$$

The set $\mathcal{A}_1(s_1)$ contains tomorrow's feasible trades. At the first date we therefore compare

$$V_0(s_0) = \max_{a_0 \in \mathcal{A}_0(s_0)} \mathbb{E}[R_0 + V_1(s_1) \mid s_0, a_0].$$

The inner decision is made after observing s_1, so its chosen trade can depend on that state. Choosing a_1 with knowledge of the later shock would solve a different, clairvoyant problem.

Key idea

If I act now, I receive X, move to a random future state, and then behave optimally from there. In these equations, X is the current reward and the value of "from there" is V_1.

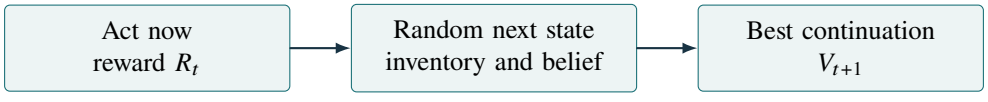


Figure 17.1: The Bellman recursion first averages over the next state and then compares current actions. Future actions depend only on information available then.

17.2 A sufficient state

Let

$$s_t = (t, O_t^{\text{elig}}, b_t, q_t^-, C_t, M_t, H_t^W, \mathcal{R}_t) \tag{17.1}$$

contain time, current eligible-contract observations, the joint parameter/state belief, incoming positions, cash, collateral, running wealth high and risk-limit state $\mathcal{R}_t$. Include pending execution orders and notices whenever they affect the next transition. The state space $\mathcal{S}$ is a measurable product space; belief components are probability measures, not necessarily finite vectors.

A state is sufficient when $\mathbb{P}(s_{t+1} \in A \mid \mathcal{G}_t, a_t) = K_t(A \mid s_t, a_t)$ for every measurable A. This is the controlled Markov property. With latent states and parameters, a posterior belief is a sufficient information state under the specified model; the observed price alone usually is not.

A static rule $w_t = \pi_{\text{stat}}(s_t)$ solves a current criterion such as (16.4). Repeating it every period does not make it dynamically optimal. Today a trade changes q_t^+, incurs cost, ties up margin and affects which tomorrow trades remain possible. These effects enter K_t, with the inventory convention $q_{t+1}^- = q_t^- + a_t$ absent contract expiry or additional fills, and wealth governed by (19.1).

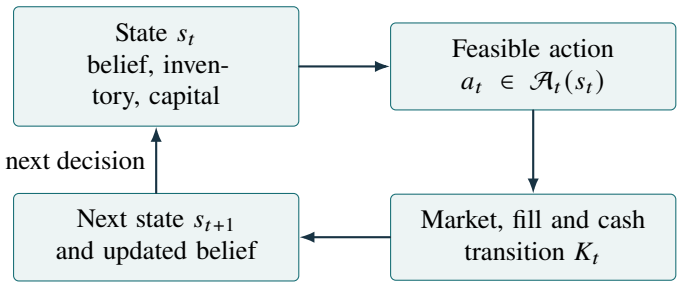


Figure 17.2: A target is not an action. The state transition uses filled trades and changes future feasibility.

17.3 The Bellman principle

The general recursion repeats the two-period argument. A transition kernel simply packages the conditional probabilities of the possible next states. The discount factor weights later rewards; it is not the risk-aversion coefficient.

A Markov decision problem is specified by $(\mathcal{S}, \mathcal{A}, K, R, \beta)$, where $\mathcal{A}_t(s)$ is a nonempty measurable feasible-action set, K_t a controlled transition kernel, $R_t(s, a, s')$ an integrable reward and $\beta \in (0, 1]$ a discount factor. Here β is distinct from risk aversion γ and from a cointegrating vector.

For a finite horizon N, terminal value $\Phi(s_N)$, and policy π,

$$J_t^\pi(s) = \mathbb{E}^\pi\left[\sum_{j=t}^{N-1} \beta^{j-t} R_j(s_j, a_j, s_{j+1}) + \beta^{N-t}\Phi(s_N) \mid s_t = s\right]. \tag{17.2}$$

Two coherent choices are possible. One uses terminal utility $\Phi = U(W_N^{\text{liq}})$ and running state-risk penalties, with trading costs already deducted from wealth. Another uses additive cost-adjusted gains and stagewise risk penalties. They are different preferences. Subtracting a cost again outside terminal utility would count it twice.

Derivation: condition on the first transition

Define $V_t(s) = \sup_\pi J_t^\pi(s)$. Condition first on the next state and separate the first action:

$$J_t^\pi(s) = \mathbb{E}^\pi[R_t(s, a_t, s_{t+1}) + \beta J_{t+1}^{\pi_{\text{tail}}}(s_{t+1}) \mid s_t = s] \tag{17.3}$$

$$\leq \sup_{a \in \mathcal{A}_t(s)} \int [R_t(s, a, s') + \beta V_{t+1}(s')] K_t(\,\mathrm{d}s' \mid s, a). \tag{17.4}$$

Conversely, choosing an optimal first action and an optimal measurable continuation attains the bound when such selectors exist; otherwise ϵ-optimal selectors approach it. Thus

$$V_N = \Phi, \qquad V_t(s) = \sup_{a \in \mathcal{A}_t(s)} Q_t(s, a), \quad Q_t(s, a) = \int [R_t + \beta V_{t+1}(s')] K_t(\,\mathrm{d}s' \mid s, a). \tag{17.5}$$

Compact feasible actions, continuous reward and weakly continuous transitions with suitable integrability are standard sufficient conditions for attainment. With finite integer action sets, measurable tie-breaking is simpler. The Bellman equation characterizes the model's best policy; it does not certify the model.

For a stationary infinite-horizon problem, bounded rewards and $\beta < 1$ make the Bellman operator a sup-norm contraction: $\|TV - TU\|_\infty \leq \beta \|V - U\|_\infty$. The inequality follows by comparing each action's expectation and then taking suprema. Iteration therefore converges to a unique bounded fixed point. Unbounded wealth utility requires additional growth conditions; finite-horizon results do not automatically establish infinite-horizon solvability [7].

17.4 A symbolic linear-quadratic control laboratory

To see a policy in algebraic form, temporarily use linear transitions and quadratic costs. Substituting a quadratic continuation value makes the current action problem another completed square; its coefficient becomes the next step of the Riccati recursion.

Let $z_t \in \mathbb{R}^d$ stack a predictable signal and inventory, with

$$z_{t+1} = F z_t + G a_t + \xi_{t+1}, \qquad \mathbb{E}[\xi_{t+1} \mid \mathcal{G}_t] = 0, \quad \operatorname{Cov}(\xi_{t+1} \mid \mathcal{G}_t) = \Omega. \tag{17.6}$$

Minimize expected cost $\mathbb{E}[\sum_{t=0}^{N-1}(z_t^\top Q_z z_t + a_t^\top R_a a_t) + z_N^\top P_N z_N]$, where $Q_z, P_N \succeq 0$ and $R_a \succ 0$. A quadratic loss on inventory minus a signal-dependent desired inventory fits $z^\top Q_z z$. This laboratory omits hard margin and integer constraints to obtain a tractable comparator.

Derivation: the Riccati recursion

Suppose the continuation cost is $z^\top P_{t+1} z + c_{t+1}$. Taking expectation after substitution yields the action-dependent terms

$$a^\top (R_a + G^\top P_{t+1} G) a + 2 a^\top G^\top P_{t+1} F z. \tag{17.7}$$

The quadratic coefficient is positive definite since $R_a \succ 0$. Completing the square gives

$$a_t^\star = -(R_a + G^\top P_{t+1} G)^{-1} G^\top P_{t+1} F z_t, \tag{17.8}$$

$$P_t = Q_z + F^\top P_{t+1} F - F^\top P_{t+1} G (R_a + G^\top P_{t+1} G)^{-1} G^\top P_{t+1} F, \tag{17.9}$$

$$c_t = c_{t+1} + \operatorname{tr}(P_{t+1} \Omega). \tag{17.10}$$

The continuation matrix changes today's trade even when today's signal is unchanged. Under partial observation, replacing z_t by its filtered mean is valid for the classical linear-Gaussian, control-independent-observation setting with quadratic cost and no hard state constraints. Margin, nonlinear costs, unknown transitions or actions that change information can break this separation.

17.5 Receding horizon and learning

Model predictive control solves a finite planning problem from today's state, executes its first action, then conditions on new information and solves again [4, 18]. A terminal approximation stands in for value beyond the planning horizon. A scenario tree imposes nonanticipativity: actions at nodes with the same observed history agree. Optimizing each complete future scenario with knowledge of its later shocks is a clairvoyant bound, not an admissible policy.

Dynamic programming uses the full continuation value; receding-horizon optimization approximates it. Neither eliminates forecast uncertainty. A terminal mean-variance objective can be time inconsistent because a variance of terminal wealth does not decompose into stagewise variances. Our base dynamic criterion uses expected terminal utility or explicit stage costs, both compatible with the stated Bellman principle.

Model-free reinforcement learning would estimate action values from experience when the transition or reward law is unavailable. If both are already specified analytically, solving their control problem directly avoids an additional unidentified approximation. Learning is relevant when actions reveal new information or the law must be learned; it requires its own exploration cost, dependence assumptions and validation.

What we gained

- A two-period policy whose continuation depends on tomorrow's information.
- A sufficient inventory-and-belief state and Bellman recursion.
- A derived linear-quadratic comparator and a receding-horizon approximation.

Where this goes next

The control problem currently speaks of trades abstractly. Execution determines how an instruction becomes actual inventory, at what cost, and with what remaining exposure during the fill interval.

Chapter 18

Execution: reaching the chosen inventory

What problem are we solving?

A target is a desired holding; an order may fill gradually or incompletely. We model the short interval between the instruction and the actual position, balancing trading cost against risk while waiting.

18.1 Two costs of changing position

Let $x(u)$ be the contracts still to acquire at execution time u. Trading rapidly makes $|\dot{x}(u)|$ large and can move the execution price against us. Trading slowly leaves more of the intended position unfilled while prices move. The objective below therefore penalizes both squared trading rate and squared remaining quantity. These terms have different units and roles.

18.2 A target is not a fill

Fix a macro-decision target $q^\star$ and current position q^0. On a short execution interval $[0, T_e]$, let $x(u) = q^\star - q(u)$ be remaining inventory to acquire. A trading rate $v(u) = -\dot{x}(u)$ satisfies $x(0) = q^\star - q^0$ and $x(T_e) = 0$. Positions may have either sign; the first derivation is scalar.

Let P^0_u be an unaffected cash-per-contract quote process with $\mathrm{d}P^0_u = \sigma_P \,\mathrm{d}B_u$, zero short-horizon drift, and constant σ_P. A stylized paid quote is $P^0_u + s\ \mathrm{sign}(v_u) + \eta v_u + \kappa_P Q_u$, where $Q_u = x(0) - x(u)$, $s \geq 0$ is half-spread, $\eta > 0$ temporary impact and $\kappa_P \geq 0$ linear permanent impact. Costs are measured against the initial unaffected quote.

Integrating traded quantity times quote difference gives expected spread and temporary cost $\int_0^{T_e} (s|v_u| + \eta v_u^2)\,\mathrm{d}u$. For absolutely continuous cumulative trades, $\int_0^{T_e} \kappa_P Q_u \dot{Q}_u \,\mathrm{d}u = \kappa_P x(0)^2/2$. This permanent-impact term is schedule-independent under this particular linear model; nonlinear or transient impact changes that conclusion.

Integration by parts yields the random shortfall $\int_0^{T_e} x(u)\,\mathrm{d}P^0_u$, up to the sign convention for acquisition versus liquidation. Its variance is $\sigma_P^2 \int_0^{T_e} x(u)^2\,\mathrm{d}u$. The mean-variance execution objective is therefore

$$\min_{x(0)=x_0,\ x(T_e)=0} \left\{ \int_0^{T_e} \left[s|\dot{x}| + \eta \dot{x}^2 + \lambda_e \sigma_P^2 x^2 \right] \mathrm{d}u + \frac{\kappa_P}{2} x_0^2 \right\}, \tag{18.1}$$

over absolutely continuous paths with square-integrable derivative, with execution-risk aversion $\lambda_e > 0$. This is the tradeoff formalized by Almgren and Chriss [1].

18.3 Deriving the schedule

We vary the entire remaining-inventory path while holding its endpoints fixed. Integration by parts moves the derivative off the perturbation; since every such perturbation must give zero first variation, the coefficient of that perturbation must vanish.

For a monotone schedule the spread integral is $s|x_0|$, independent of its shape. For a perturbation $x + \epsilon h$, where $h(0) = h(T_e) = 0$, differentiating the remaining integral gives

$$2\int_0^{T_e} [\eta \dot{x}\dot{h} + \lambda_e \sigma_P^2 x h]\,\mathrm{d}u = 2\int_0^{T_e} [-\eta \ddot{x} + \lambda_e \sigma_P^2 x]h\,\mathrm{d}u. \tag{18.2}$$

Thus the Euler equation is $\ddot{x} = \omega_e^2 x$, with $\omega_e = \sqrt{\lambda_e \sigma_P^2/\eta}$. The endpoint conditions give

$$x(u) = x_0 \frac{\sinh(\omega_e(T_e - u))}{\sinh(\omega_e T_e)}. \tag{18.3}$$

The integral is strictly convex in the path, so this stationary path is the unique minimizer. At $\lambda_e \downarrow 0$, the ratio tends to $(T_e - u)/T_e$, yielding constant-rate trading.

Trading a fixed quantity over a vanishing interval sends temporary cost upward like x_0^2/T_e. Spreading a fixed-shape schedule over an indefinitely long interval raises its unfilled-inventory variance like T_e. Signal decay or adverse drift would create an additional opportunity-cost term. Those modifications should use the same predictive law as the allocation decision.

In multiple contracts, replace $\eta \dot{x}^2$ by $\dot{x}^\top \Lambda_e \dot{x}$, $\Lambda_e \succ 0$, and $\sigma_P^2 x^2$ by $x^\top \Omega_e x$, $\Omega_e \succeq 0$. Then $\Lambda_e \ddot{x} = \lambda_e \Omega_e x$. Cross-contract risk couples schedules. Rate and liquidity bounds give a constrained boundary-value problem without this general closed form.

18.4 How execution returns to allocation

Let $\mathcal{E}_t(q^-, q^\star)$ be the minimized expected execution cost plus any separately stated execution-risk penalty. Allocation can penalize $\mathcal{E}_t$, while the ledger deducts realized cash cost only. A risk penalty is not an expense. If a cost coefficient in the portfolio objective already includes expected spread and impact, those components are replaced by $\mathcal{E}_t$, not added again.

With uncertain fills, the control is an order instruction and actual a_t is a random fill with a conditional law depending on instruction and market state. The feasible policy constrains intermediate inventories, not only the final target. If liquidity disappears, the target may never be reached; subsequent market gains use the filled position.

What we gained

- An instruction, a fill process and an actual inventory distinguished.
- An execution objective derived from impact and price uncertainty.
- An optimal scalar schedule and its feedback into allocation.

Where this goes next

We can now reconcile the whole experiment in cash: settlement gains use filled holdings, costs use actual executions, and collateral transfers must not be mistaken for losses.

Chapter 19

Exact cash accounting and attribution

What problem are we solving?

Forecasts and execution models can be wrong. The account must still balance. We separate the exact wealth transition from optional explanations of which factor or carry component generated the observed gain.

19.1 Separate an internal transfer from an expense

Write $W = C + M$, where C is free cash and M is collateral. Moving an amount u from cash to collateral changes these to $C - u, M + u$, whose sum is unchanged. Paying a fee reduces the sum. This elementary distinction determines every sign in the full ledger.

After trading, inventory is the incoming position plus actual fills. Interest accrues on the resulting cash balances; settlement gains accrue to the resulting inventory. We now write those steps explicitly.

19.2 Self-financing with collateral

Let q_t^- be incoming inventory and a_t actual filled trades, so $q_t^+ = q_t^- + a_t$. Let $W_t = C_t + M_t$ be account equity just before trading, partitioned into available cash C_t and collateral M_t. Futures have zero marked value immediately after settlement; an unsettled variation-margin receivable, if present, must be added to W_t.

Let $K_t(a_t) \geq 0$ be cash trading cost and let R_t^C, R_t^M be one-period cash and collateral gross accrual factors. Choose posted collateral $M_t^+ \geq \mathfrak{m}_t(q_t^+)$, where $\mathfrak{m}_t$ is the contemporaneous margin requirement. Available cash after trades is $C_t^+ = W_t - K_t(a_t) - M_t^+$. With $H_t^{\text{fund}} \geq 0$ denoting additional borrowing or holding charges,

$$\boxed{W_{t+1} = R_t^C C_t^+ + R_t^M M_t^+ + (q_t^+)^\top D_t r_{t+1} - H_t^{\text{fund}}.} \tag{19.1}$$

Moving cash into margin does not change equity. The cost term is deducted once, before accrual. If cash and collateral share R_t^f, then

$$W_{t+1} - W_t = (R_t^f - 1)W_t + (q_t^+)^\top D_t r_{t+1} - R_t^f K_t(a_t) - H_t^{\text{fund}}. \tag{19.2}$$

The excess gain over holding initial capital in the cash account is the last three terms. There is no condition $\sum_i w_i = 1$ for futures exposures.

A self-financing policy has no external deposit or withdrawal in (19.1). Margin replenishment transfers internal cash; if insufficient, a specified liquidation transition applies. An admissible strategy is predictable in the observable information, obeys position and delivery restrictions, and satisfies $\mathbb{E}\sum_t \|D_t q_t^+\|^2 \|r_{t+1}\|^2 < \infty$, finite expected costs, and a prescribed lower bound on discounted wealth up to its stopping time. Such restrictions rule out doubling schemes.

A Gaussian forecast has unbounded losses, so no finite margin reserve guarantees pathwise solvency for nonzero positions. We impose an initial feasible reserve and a declared liquidation/default state when later cash is insufficient; limited default loss is not silently assumed. A bounded scenario stress constraint is protection on that scenario set only.

19.3 Accounting and attribution

An attribution partitions a gain already present in the account. Projecting market gains into common and residual components does not generate additional revenue. We first sum actual holdings times price increments, then split that same sum.

On a fine grid indexed by j, let q_j^+ be the inventory actually held during the next increment, and let F_j be the matching vector of contract prices. With multiplier matrix $M = \mathrm{diag}(m_i)$, exact market P&L is $\sum_j (q_j^+)^\top M(F_{j+1} - F_j)$. There is no substitution of target inventory for actual inventory.

For one macro interval, suppose normalization is fixed and define $w_j = D_t q_j^+ / A_t$. The coordinate identity gives

$$\frac{\Pi_{\text{market}}}{A_t} = \sum_j w_j^\top \bar{r}_j + \sum_j (V_k^\top w_j)^\top f_{j+1} + \sum_j (U^\top w_j)^\top e_{j+1}, \tag{19.3}$$

where $\bar{r}_j$ is the centering contribution for the fine interval and the coordinate gains are defined consistently. This is a factor/residual attribution of the *same* market P&L. A realized residual contribution is not proof of alpha; its conditional predictable part and innovation can be further separated.

An alternative economic partition first writes $r_{j+1} = c_j + \widetilde{r}_{j+1}$, where c_j is a declared predictable carry/roll-down attribution derived from a curve model. Then project only $\widetilde{r}$:

$$\Pi_{\text{market}} = A_t \sum_j w_j^\top c_j + A_t \sum_j w_j^\top P_k \widetilde{r}_{j+1} + A_t \sum_j w_j^\top P_k^\perp \widetilde{r}_{j+1}. \tag{19.4}$$

Adding carry to a factor/residual split of the full return would double count it. Changing the carry convention changes this attribution, but not total market gain. Splice gaps from (2.4) are excluded; explicit roll trades contribute costs and new exposures.

The balanced equity identity, allowing different accrual rates, is

$$\begin{aligned} W_{t+1} - W_t = \Pi_{\text{market}} &+ (R_t^C - 1)C_t^+ + (R_t^M - 1)M_t^+ \\ &- K_t^{\text{spread}} - K_t^{\text{temp}} - K_t^{\text{perm}} - K_t^{\text{fees}} - H_t^{\text{fund}}. \end{aligned} \tag{19.5}$$

Here K_t is the sum of its displayed components and $C_t^+ + M_t^+ = W_t - K_t$. If paid quotes already incorporate impact, market P&L must be measured relative to the same unaffected benchmark before these impact terms are subtracted. Using transaction prices in gains and subtracting the same slippage again is incorrect.

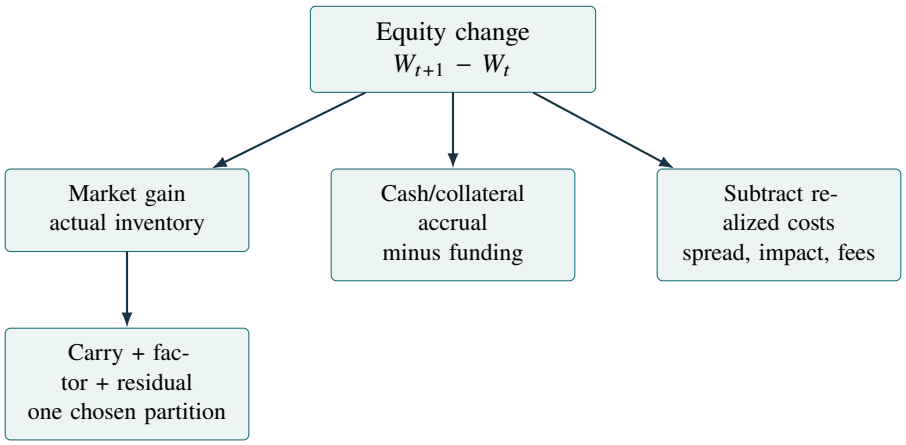


Figure 19.1: P&L attribution respects a cash identity. Carry and factor labels partition market gains; they are not extra revenue streams.

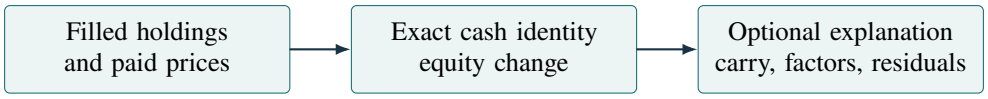


Figure 19.2: The account balances before any statistical interpretation is attached to the gain.

What we gained

- A self-financing wealth recursion including collateral and funding.
- A gain sum based on actual holdings and identified contracts.
- An attribution convention that cannot count carry or costs twice.

Where this goes next

A balanced ledger tells us what happened, not whether the forecast helped. We finish the main argument by specifying evidence that can distinguish a useful model from noise and selection.

Chapter 20

Testing whether the system works

What problem are we solving?

A sophisticated model can fit history without improving any future decision. We distinguish predictive information, economic usefulness and safe operation, then specify tests that are allowed to reject each claim.

20.1 Three questions to keep separate

First ask whether the proposed residual observation changes a future gain distribution beyond benchmark information. Next ask whether a policy using that change improves the declared economic criterion after costs. Finally ask whether the model and account remain within the operating conditions under which either claim was evaluated. Good covariance reconstruction answers none of these alone.

20.2 Two hypotheses, not one

Let $r_{t+1} \in L^2(\mathbb{P}; \mathbb{R}^n)$ be the normalized gain vector defined in Chapter 1. Let $\mathcal{G}_t^0 \subseteq \mathcal{G}_t$ contain benchmark information, including past common factors and admissibility variables. Let x_t^{obs} be an observable residual-state statistic, and write $\mathcal{G}_t^1 = \sigma(\mathcal{G}_t^0, x_t^{\mathrm{obs}}) \subseteq \mathcal{G}_t$. Our predictive question is

$$H_0^{\mathrm{pred}}: \quad \mathcal{L}_{\mathbb{P}}(r_{t+1} \mid \mathcal{G}_t^1) = \mathcal{L}_{\mathbb{P}}(r_{t+1} \mid \mathcal{G}_t^0) \quad \mathbb{P}\text{-almost surely}, \tag{20.1}$$

$$H_1^{\mathrm{pred}}: \quad \mathbb{P}\Big(\mathcal{L}_{\mathbb{P}}(r_{t+1} \mid \mathcal{G}_t^1) \neq \mathcal{L}_{\mathbb{P}}(r_{t+1} \mid \mathcal{G}_t^0)\Big) > 0. \tag{20.2}$$

Equality of laws means equality of conditional probabilities for every Borel set. It is stronger than equality of conditional means. A change only in tail risk can matter to a decision even if expected gains are unchanged.

For a predeclared feasible policy family Π, define G_{t+1}^{π} as its wealth gain net of costs and benchmark collateral return. An economic hypothesis is

$$H_0^{\mathrm{econ}}: \quad \sup_{\pi \in \Pi} \mathbb{E}[G_{t+1}^{\pi}] \leq \delta_{\mathrm{econ}}, \qquad H_1^{\mathrm{econ}}: \quad \sup_{\pi \in \Pi} \mathbb{E}[G_{t+1}^{\pi}] > \delta_{\mathrm{econ}}, \tag{20.3}$$

where $\delta_{\mathrm{econ}} \geq 0$ is a minimum economically relevant gain in the declared capital units. Risk-adjusted utility comparisons use the separately declared preference, rather than this mean-gain null. Positive mean compensation for bearing common risk does not establish conditional relative-value skill.

A factor hypothesis is yet another statement: for some $k < n$, a rank-k projector P satisfies $\mathbb{E}\|(I-P)(r-\mathbb{E}r)\|^2 \le \epsilon_{\text{fac}}\mathbb{E}\|r-\mathbb{E}r\|^2$, where $\epsilon_{\text{fac}} \in (0,1)$ is chosen before evaluation. This statement neither implies H_1^{pred} nor stationarity of cumulative residuals.

20.3 Seal the research experiment

Before evaluation fix the eligible contract families, information release rules, normalization, rolling rule, training lengths, candidate state models, factor-dimension candidates, cost assumptions, ambiguity radii, feasible risk limits, policy objective, benchmark and test family. Let $\mathcal{T}_{\text{train}}$, $\mathcal{T}_{\text{select}}$, and $\mathcal{T}_{\text{eval}}$ be chronologically ordered disjoint blocks. Select models on the middle block and evaluate the locked selection procedure on the last.

For rolling estimation, each estimate at t is a map of $\{O_s : s \le t\}$. For expanding estimation, the history grows rather than replacing older observations. Neither convention permits using future evaluation outcomes to choose historical transformations. If a target spans $[t, t+h]$, a training target is available only after $t+h$; purge boundary targets whose outcome intervals overlap an evaluation block.

Declare a no-residual-information benchmark with the same contract eligibility, costs, capital and risk constraints. Otherwise the comparison can measure different leverage or trading intensity rather than value added by the residual state. Retain every candidate considered when defining the multiplicity family.

20.4 Forecast errors and proper scoring

A mean forecast is assessed by conditional errors; a full distribution needs a score sensitive to the whole law. The log-score calculation below explains why concentrating probability on outcomes that actually occur is rewarded in expectation.

For a conditional mean forecast $\widehat{\mu}_t$, error is $u_{t+1} = r_{t+1} - \widehat{\mu}_t$. If the model is correctly specified, $\mathbb{E}[u_{t+1} \mid \mathcal{G}_t] = 0$. This implies moment restrictions $\mathbb{E}[h(\mathcal{G}_t)^\top u_{t+1}] = 0$ for preselected square-integrable instruments h. Zero average error alone does not establish the conditional restriction.

For a predictive density p_t, negative log score is $-\log p_t(r_{t+1})$. If the true density is p_t°, conditional expected score difference equals $\int p_t^\circ \log(p_t^\circ/p_t)$, a Kullback–Leibler divergence. Jensen's inequality applied to $\log(p_t/p_t^\circ)$ gives nonnegativity when the densities have compatible support and finite integrals. Thus expected log score rewards the full law. It may be unsuitable when tail logs are not integrable or densities assign zero mass to realized observations.

For a continuous correctly specified scalar conditional distribution F_t, the transform $U_{t+1} = F_t(Y_{t+1})$ is conditionally uniform: for $u \in (0,1)$, $\mathbb{P}(U_{t+1} \le u \mid \mathcal{G}_t) = F_t(F_t^{-1}(u)) = u$. At atoms use a randomized distributional transform with independent auxiliary uniform randomness. This randomization concerns a diagnostic, not a trading action. Conditional uniformity and serial-independence checks test different failures.

20.5 Dependence and economic significance

For independent observations, the variance of a sample average is marginal variance divided by sample size. Expanding the variance of a sum of dependent gains also produces all lag covariances. Their contribution is the long-run variance in the next calculation.

Let D_t be a predeclared scalar score improvement or net gain difference between the chosen policy and benchmark, both evaluated with the same capital convention. Under stationarity, suitable

strong mixing, finite $2+\epsilon$ moments, and a valid central limit theorem,

$$\sqrt{N}(\overline{D}_N - \mathbb{E}D_t) \Rightarrow \mathcal{N}(0, \omega^2), \qquad \omega^2 = \sum_{\ell=-\infty}^{\infty} \mathrm{Cov}(D_t, D_{t-\ell}) > 0. \tag{20.4}$$

The covariance sum must converge. Estimate it by

$$\widehat{\omega}_N^2 = \widehat{\gamma}_0 + 2\sum_{\ell=1}^{L_N}\left(1 - \frac{\ell}{L_N+1}\right)\widehat{\gamma}_\ell, \tag{20.5}$$

where $\widehat{\gamma}_\ell = N^{-1}\sum_{t=\ell+1}^{N}(D_t - \overline{D}_N)(D_{t-\ell} - \overline{D}_N)$. The Bartlett weighting gives the familiar positive-semidefinite construction [15]. Consistency additionally needs bandwidth growth slow enough for the dependence and moment assumptions; $L_N \to \infty$, $L_N/N \to 0$ alone is not a universal theorem.

Test $H_0 : \mathbb{E}D_t \le \delta_{\mathrm{econ}}$ with $T_N = \sqrt{N}(\overline{D}_N - \delta_{\mathrm{econ}})/\widehat{\omega}_N$. Reject only if T_N exceeds the preselected one-sided critical value under the maintained limit law. Economic significance requires exceeding δ_{econ} after costs, not just exceeding zero before costs. Overlapping h-period targets share shocks, so observations cannot be treated as independent merely because decision dates differ.

If the dependence assumptions fail, the nominal critical value loses its justification. A block resampling procedure can preserve short-range dependence under an appropriate block-length asymptotic regime, but cannot make structural breaks disappear. A finite-sample martingale test is another possibility only after imposing the corresponding conditional moment and tail bounds.

20.6 Selection, breaks and rejection criteria

For M predeclared hypotheses with valid marginal p-values, Bonferroni rejects each at level α/M and controls familywise error by the union bound, regardless of cross-test dependence. More powerful joint procedures can account for the distribution of the maximum selected performance statistic; White [19] addresses this data-snooping problem. Testing only the final winner as if it had been specified initially does not.

Repeated looks at evaluation outcomes also create selection. Use a fixed evaluation horizon, or preallocate levels α_j across looks with $\sum_j \alpha_j \le \alpha$. Optional stopping without such a rule invalidates a fixed-horizon claim. Failure to reject the null is not proof that the null is true.

The experiment has the following explicit decisions. Reject a maintained transition model if its predeclared conditional-moment or distribution-calibration test rejects at its adjusted level. Reject the useful-edge hypothesis if an upper confidence bound for net gain improvement lies below δ_{econ}; merely failing to demonstrate superiority is recorded as inconclusive. Suspend allocation when the feasible set is empty, model inputs violate their defined domain, cash reserve fails, or predeclared principal-angle and covariance-conditioning limits are exceeded. Structural-break diagnostics compare recent predictive scores and moments with their training restrictions under a declared sequential error budget.

Costs and capacity require sensitivity bounds. Let $\mathcal{K}$ be a declared set of plausible cost maps; report whether the lower confidence bound on improvement remains positive for every $K \in \mathcal{K}$. A signal whose economic conclusion reverses under a small admissible cost perturbation is not robustly supported. No empirical outcome is assumed here.

20.7 Why these models are connected

Having derived the chain, we can now locate the research literature within it. These sources justify mathematical tools and motivate hypotheses; they do not supply an empirical result for the constructed futures system.

The factor-residual research question is motivated by Avellaneda and Lee [2], whose equity construction is not evidence that the same relation holds in futures. We retain its separation of common variation from residual structure, but define a different gain ledger and treat residual stability as testable. PCA supplies an approximation criterion [11]; cointegration instead restricts level processes [6, 10]. Neither can replace the other merely because both use linear combinations.

Mean-variance allocation [14] is the first tractable decision laboratory. Shrinkage [13] addresses the instability of the risk input, while robust optimization [8] changes how uncertain expected gains are used. The optimization and duality arguments follow the finite-dimensional framework of Boyd and Vandenberghe [3]. Inventory couples decisions through time, making the multi-period formulations of Boyd et al. [4], Skaf and Boyd [18] relevant. Execution then introduces the separate time scale studied by Almgren and Chriss [1].

The main statistical specification is a stable linear latent residual-return state in a factor basis fixed over each research block. It does not require a stationary cumulative price residual. A Gaussian version permits exact filtering; heavy-tailed and regime versions are challengers. This choice is provisional: predictive calibration, cost-adjusted utility, subspace stability and cash feasibility can each invalidate it.

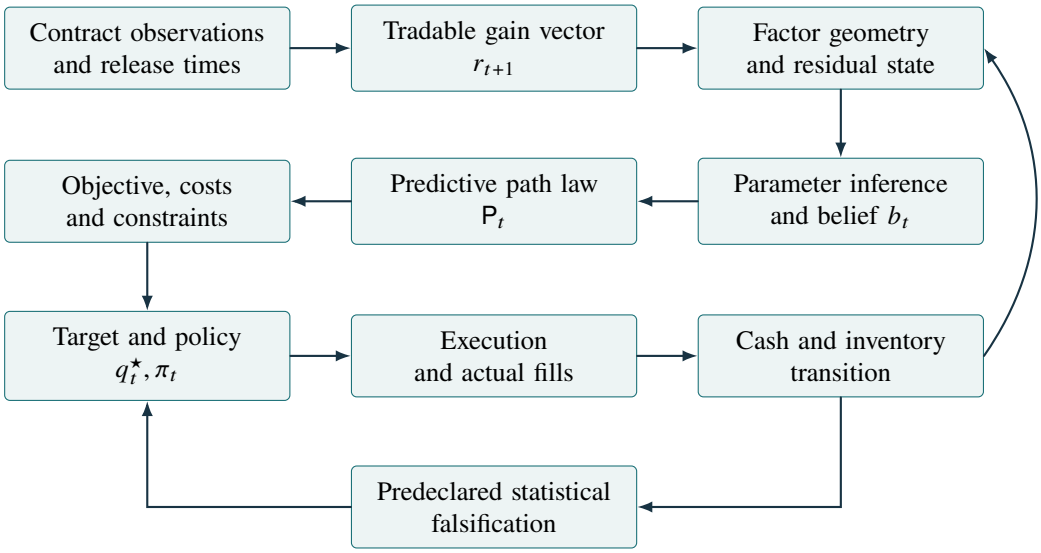


Figure 20.1: The dependency graph. The return arrow carries next-period observations and inventory, never future information into the current decision.

Mathematical checkpoint

Could a covariance be almost perfectly low rank while every admissible residual strategy has nonpositive expected gain after costs? Which part of the research contract would survive?

What we gained

- Separate predictive and economic hypotheses with matched benchmarks.
- Chronological validation, proper scores and dependence-aware uncertainty.
- Predeclared rejection, suspension and multiple-testing rules.

Where this goes next

The mathematical pieces are now available. The appendices retain the technical qualifications. The closing two-page sheet keeps an observed-state, one-period baseline and adds a slow thesis state, joint path forecasts, multi-period control and a separate execution controller.

Appendix A

Probability and information: technical foundations

What problem are we solving?

Conditional averages and release-time rules were introduced through finite examples. Here we state their general probability-space meaning and explain why the same definitions also cover hidden states.

A.1 Probability and information

Work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_u)_{u\geq 0}, \mathbb{P})$ satisfying the usual conditions: completeness and right continuity. $\mathbb{P}$ describes realized market uncertainty, not risk-neutral valuation. Trading dates $0 = t_0 < t_1 < \cdots < t_N$ are deterministic for the base construction. We abbreviate the date index t_j to j outside continuous-time sections.

The observable filtration is

$$\mathcal{G}_t = \sigma\{O_s : \text{observation } O_s \text{ was released by } t\} \vee \sigma\{\text{own fills and cash notices received by } t\} \subseteq \mathcal{F}_t. \tag{A.1}$$

An unobserved market state may be $\mathcal{F}_t$-measurable without being $\mathcal{G}_t$-measurable. Revised observations are new observations at their release dates. Estimated parameters, contract-selection rules, targets and orders at t must all be $\mathcal{G}_t$-measurable. In particular a statistic fit through $t + 1$ cannot be used in a_t.

Conditional expectation $\mathbb{E}[Y \mid \mathcal{G}_t]$ is the $\mathcal{G}_t$-measurable integrable random variable satisfying $\mathbb{E}[\mathbf{1}_A \mathbb{E}(Y \mid \mathcal{G}_t)] = \mathbb{E}[\mathbf{1}_A Y]$ for every $A \in \mathcal{G}_t$. For square-integrable Y, it minimizes $\mathbb{E}[(Y - Z)^2]$ over $\mathcal{G}_t$-measurable Z; expand the square and use the vanishing conditional cross term. This projection meaning will reappear in filtering, whereas PCA projects in an asset-coordinate space.

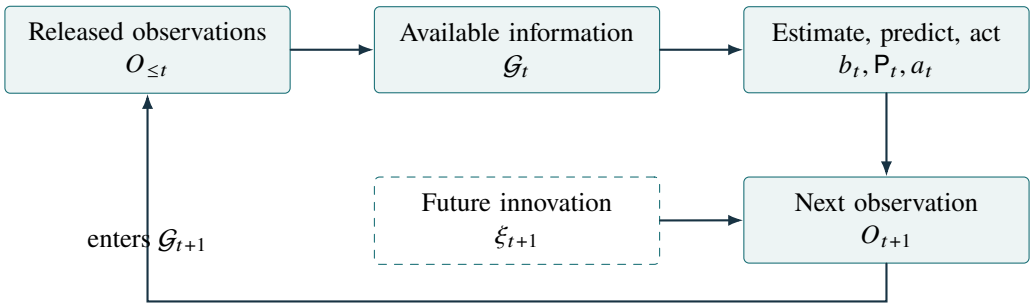


Figure A.1: Information flow. No arrow from the future innovation enters the current belief or action.

What we gained

- An observable filtration distinct from latent full information.
- Conditional expectation defined by its integral identity.
- Its least-squares projection interpretation.

Where this goes next

The next appendix verifies projection identities and the sensitivity of estimated factor spaces.

Appendix B

Projection and subspace perturbation details

What problem are we solving?

The main text used orthogonality and eigengaps geometrically. We now prove the identities and compute the first-order effect of a covariance perturbation.

B.1 Projection identities

If $V^\top V = I_k$, then $P = VV^\top$ satisfies $P^2 = V(V^\top V)V^\top = P$. Symmetry gives $\langle Pz, (I-P)z\rangle = z^\top(P - P^2)z = 0$. Consequently $\|z\|^2 = \|Pz\|^2 + \|(I-P)z\|^2$. For any $a \in \mathbb{R}^k$,

$$\|z - Va\|^2 = \|(I-P)z\|^2 + \|V^\top z - a\|^2. \tag{B.1}$$

The least-squares coordinate is $a = V^\top z$. This proves both projection and reconstruction claims without an unexplained eigendecomposition.

B.2 Why an eigengap appears

Let $\Sigma v_i = \lambda_i v_i$ with distinct λ_i, and perturb to $(\Sigma + \epsilon E)(v_i + \epsilon h_i) = (\lambda_i + \epsilon\dot{\lambda}_i)(v_i + \epsilon h_i) + o(\epsilon)$. Normalization imposes $v_i^\top h_i = 0$. Matching first-order terms gives $(\Sigma - \lambda_i I)h_i = -(E - \dot{\lambda}_i I)v_i$. Premultiplication by $v_i^\top$ gives $\dot{\lambda}_i = v_i^\top E v_i$; premultiplication by $v_j^\top$, $j \neq i$, gives

$$v_j^\top h_i = \frac{v_j^\top E v_i}{\lambda_i - \lambda_j}. \tag{B.2}$$

Mixing between directions inside a retained cluster changes its basis but need not change its projector. Mixing across the retained/discarded boundary is controlled by the boundary gap. When that gap closes, differentiability of the selected projector can fail, so first-order reasoning is no longer valid.

What we gained

- A direct proof of orthogonal reconstruction.
- Eigenvalue and eigenvector perturbation equations.
- The role of separation between retained and discarded directions.

Where this goes next

The next appendix checks the inference identities used when propagating state and parameter uncertainty.

Appendix C

Inference identities and uncertainty

What problem are we solving?

Gaussian updating and stable covariances were derived in the main text. The following calculations provide alternative algebraic checks and show exactly where parameter constraints break conjugacy.

C.1 Technical note: the stationary AR construction

The stationary causal solution is $X_t = \theta + \sum_{j=0}^{\infty} b^j \epsilon_{t-j}$. The series converges in L^2 because $\sum_j b^{2j} < \infty$. It is strictly stationary for identically distributed two-sided innovations, and, as a measurable causal transform of an independent sequence, is ergodic. Its Gaussian stationary law is $\mathcal{N}(\theta, \sigma_\epsilon^2/(1-b^2))$. Without Gaussianity its characteristic function is $e^{iu\theta}\prod_{j\geq 0}\varphi_\epsilon(b^j u)$, with $u \in \mathbb{R}$ and $\varphi_\epsilon(u) = \mathbb{E}[e^{iu\epsilon}]$. When $|b| \geq 1$, nondegenerate innovations prevent a stable finite-variance causal stationary solution. Negative b yields alternating decay; $b = 1$ with $a = 0$ gives a random walk.

C.2 A conjugate regression calculation

Consider an observed regression $Y = Z\beta + \epsilon$, with $\epsilon \sim \mathcal{N}(0, \sigma^2 I_N)$, known $\sigma^2 > 0$, $\beta \in \mathbb{R}^p$, and prior $\beta \sim \mathcal{N}(b_0, H_0)$, $H_0 \succ 0$. Multiplying likelihood and prior gives a log density, up to constants,

$$-\frac{1}{2}\left[\beta^\top(H_0^{-1} + \sigma^{-2}Z^\top Z)\beta - 2\beta^\top(H_0^{-1}b_0 + \sigma^{-2}Z^\top Y)\right]. \tag{C.1}$$

Complete the square:

$$H_N = (H_0^{-1} + \sigma^{-2}Z^\top Z)^{-1}, \quad b_N = H_N(H_0^{-1}b_0 + \sigma^{-2}Z^\top Y). \tag{C.2}$$

The prior precision ensures invertibility even with deficient Z, but then inference in unsupported directions is prior-driven. For future regressor $z_\star$, integrating β gives $Y_\star \mid Y \sim \mathcal{N}(z_\star^\top b_N, \sigma^2 + z_\star^\top H_N z_\star)$. The second variance term is precisely coefficient uncertainty.

An unconstrained Gaussian coefficient prior assigns mass to unstable AR transitions. If stability is required, truncate the prior to $|b| < 1$ or $\rho(A) < 1$; conjugate Gaussian expressions no longer describe the exact posterior. Numerical integration over the constrained posterior is then the stated mathematical problem.

C.3 OU covariance and stationarity

For stationary scalar OU with variance $\sigma^2/(2\kappa)$, (8.3) gives $\mathrm{Cov}(X_{t+h}, X_t) = e^{-\kappa h}\,\mathrm{Var}(X_t)$ for $h \geq 0$. The future stochastic integral is independent of X_t, which eliminates the cross term. This covariance restriction is testable across lags and sampling intervals.

For a vector transition $x_{t+1} = Ax_t + w_{t+1}$, a stationary covariance satisfies $\mathrm{vec}(P) = (A \otimes A)\,\mathrm{vec}(P) + \mathrm{vec}(Q)$. Here vec stacks matrix columns and $\otimes$ is the Kronecker product. If $\rho(A) < 1$, every eigenvalue product of $A \otimes A$ has magnitude below one; hence $I - A \otimes A$ is invertible. The convergent covariance series is usually more informative than this inverse expression because it shows positivity.

C.4 Kalman update as completion of a square

When $P^- \succ 0$ and $R \succ 0$, the posterior exponent is

$$(x - m^-)^\top (P^-)^{-1}(x - m^-) + (y - Cx)^\top R^{-1}(y - Cx). \tag{C.3}$$

Collecting quadratic and linear terms gives posterior precision $(P^-)^{-1} + C^\top R^{-1} C$ and precision-weighted mean $(P^-)^{-1}m^- + C^\top R^{-1} y$. The inverse precision equals $P^- - P^- C^\top (CP^- C^\top + R)^{-1} CP^-$: multiply this candidate by the precision matrix and cancel terms using $CP^- C^\top = S - R$. Substituting the candidate into the mean yields (10.5). The covariance-completion proof in the main text also covers singular P^-.

What we gained

- A conjugate regression posterior and predictive variance.
- OU lag covariance and the vectorized Lyapunov equation.
- The information-form Kalman update and its domain.

Where this goes next

The final appendix examines policy randomization, constraint value and conservation without interrupting the main trading argument.

Appendix D

Policy qualifications and accounting checks

What problem are we solving?

A distribution over unknown parameters differs from a distribution over chosen actions. We keep this advanced distinction with the duality, control and ledger checks that delimit the main constructions.

D.1 Deterministic, Bayesian and stochastic policies

A deterministic policy chooses $a_t = \pi_t(s_t)$. A stochastic policy uses a probability kernel $\pi_t(\mathrm{d}a \mid s_t)$, so the realized action is drawn conditionally on s_t. Given Q_t, $\int Q_t(s,a)\pi_t(\mathrm{d}a \mid s) \le \sup_a Q_t(s,a)$. Thus without another criterion, randomization cannot improve a known-state unconstrained expected-value maximization. It can matter with exploration, strategic interaction, or special distributional constraints, but must be justified.

A Bayesian action can be deterministic even when its state is an entire posterior distribution. Parameter uncertainty has already been integrated into Q_t. Smaller, more robust or zero positions are possible responses to uncertainty; random actions are not a substitute for valid beliefs.

For a finite feasible action set and temperature $\tau > 0$, consider

$$\max_{p_a \ge 0,\ \sum_a p_a = 1} \left\{ \sum_a p_a Q_t(s,a) - \tau \sum_a p_a \log p_a \right\}. \tag{D.1}$$

Differentiating the Lagrangian gives $Q_a - \tau(1 + \log p_a) - \lambda = 0$. Normalization yields

$$p_a^\star = \frac{\exp(Q_a/\tau)}{\sum_b \exp(Q_b/\tau)}, \qquad V_t^\tau(s) = \tau \log \sum_a \exp(Q_t^\tau(s,a)/\tau). \tag{D.2}$$

This is the Bellman recursion for $\mathbb{E}_\pi \sum_t \beta^t [R_t + \tau \mathcal{H}(\pi_t(\cdot \mid s_t))]$, with Shannon entropy $\mathcal{H}(p) = -\sum_a p_a \log p_a$. Factoring out $\exp(\max_a Q_a/\tau)$ proves that $V^\tau \to \max Q$ and mass concentrates on maximizing actions as $\tau \downarrow 0$. Ties can retain a nondegenerate limiting mixture.

For continuous actions, entropy depends on the reference measure and units. A defensible alternative is $-\tau\,\mathrm{KL}(\pi \| \pi_0)$ relative to a specified reference policy π_0, giving density proportional to $e^{Q/\tau}$ relative to π_0, provided its integral is finite. Entropy rewards dispersion by design; it is not a free improvement in trading utility.

D.2 Equality-constrained loss of value

Put $d = Ew_0 - b^E$ and $H = E\Sigma^{-1}E^\top \succ 0$. The adjustment in (15.1) is $\Delta w = -\Sigma^{-1}E^\top H^{-1}d$. Therefore

$$J(w_0) - J(w^\star) = \frac{\gamma}{2}\Delta w^\top \Sigma \Delta w = \frac{\gamma}{2} d^\top H^{-1} d. \tag{D.3}$$

This is the criterion cost of the specified equalities, not an observed financial loss. It grows when an expensive exposure constraint conflicts with the unconstrained mean-risk preference.

D.3 A Bellman accounting convention

Suppose terminal utility is used and costs are deducted in $\mathcal{W}$. Then the Bellman integrand is $-\psi_t(s,a) + \beta V_{t+1}(\mathcal{W}(s,a,\xi))$, where ψ_t is a noncash risk or control penalty. If instead rewards are net normalized cash increments, terminal value must be chosen so those increments are not counted again as terminal wealth. Discounting a cash gain and discounting utility are also distinct conventions.

The finite-horizon Riccati inverse in Chapter 17 exists because $u^\top(R_a + G^\top P_{t+1} G)u \geq u^\top R_a u > 0$. Positive semidefiniteness of P_t follows because $z^\top P_t z$ is the minimized nonnegative deterministic quadratic cost-to-go. Infinite-horizon convergence requires stabilizability/detectability conditions beyond this finite-horizon argument.

D.4 Four conservation checks

With zero position, zero costs, no extra funding charge and equal rates, (19.1) reduces to $W_{t+1} = R_t^f W_t$. With zero rates and unchanged futures quotes, trading reduces wealth exactly by its cost. Moving a cash amount m from C to M leaves $W = C + M$ unchanged at the transfer time. Switching maturities at unchanged quotes creates neither premium revenue nor the splice gain in (2.4).

Finally, for $r = c + P(r - c) + P^\perp(r - c)$, multiplication by actual dollar exposures proves (19.4) exactly. This identity survives a poor carry model, poor factor model or poor forecast. Those failures change the interpretation and decision quality, while the accounting equality remains the audit anchor.

D.5 Fixing a policy within the family

For one fully declared baseline, fix the stable Gaussian factor/residual model, frozen-block observation coordinates, a proper parameter prior supported on the stability and covariance restrictions, and the corresponding exact belief recursion. Centering error may be included as an unknown constant intercept with its own prior; setting it to zero is an additional maintained restriction. Choose bounded contract instructions and a specified fill law, including immediate full fills as a transparent limiting case.

Fix reference capital $A_0 > 0$ across the finite decision horizon. Let $G_t(s,a,s') = [W_{t+1} - R_t^f W_t]/A_0$ be the next realized excess gain from the exact ledger, including actual trading costs. A dynamically consistent criterion is the additive stage reward

$$R_t(s,a,s') = G_t(s,a,s') - \frac{\gamma}{2}\operatorname{Var}^{\mathbb{P}}(G_t \mid s,a) - \eta\|a\|_1, \qquad \Phi(s_N) = -K_N^{\mathrm{liq}}/A_0. \tag{D.4}$$

Here $\gamma > 0$, $\eta \geq 0$, and K_N^{liq} is the remaining terminal liquidation cost, not already charged in G_{N-1}. The stage variance is a conditional model quantity known at the decision; it is not a realized cash debit. Set $\beta = 1$ for this finite-horizon excess-gain preference and impose the margin, tail and

factor-tolerance restrictions at each state. The Bellman maximizing selector with a fixed tie-break rule is then the baseline policy.

The one-period portfolio objective is its myopic approximation; a horizon optimizer with an approximate terminal value is its receding-horizon approximation. They need not select the same positions. Mean ambiguity can replace the stage conditional mean by its support-function bound. If ambiguity also changes transition laws, time-consistent worst-case Bellman recursion requires conditional ambiguity sets stable under pasting; a single unknown model fixed for the entire path is instead carried in the belief. This specifies which notion of optimality the final action uses.

What we gained

- A derivation of entropy-regularized policies and their changed objective.
- The objective cost of equality constraints.
- Finite-horizon control and cash-conservation checks.

Where this goes next

Return to the closing operational sheet with these qualifications in mind, or proceed to the options volume to study a gain map that includes a nonlinear pricing problem.

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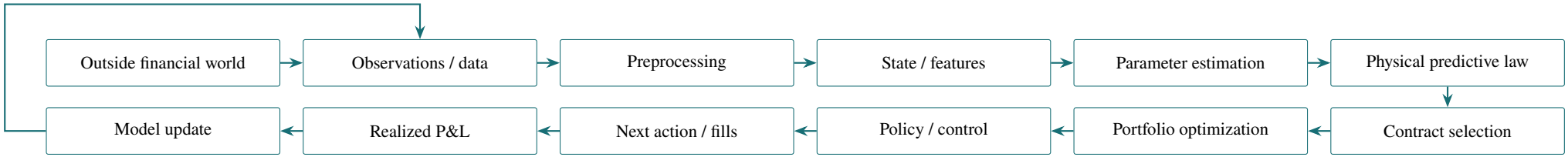
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Multi-asset futures: data, thesis and path law

One-period baseline retained; optional slow thesis state and multi-period control use the same gain and cash maps.



Observations update beliefs; only decision dates (or mandatory risk events) trigger strategic orders. Execution has its own faster clock.

Three clocks. Observations arrive at o_j; decisions occur at t_k with spacing Δ_d; the economic thesis expires at a fixed T^{th} (e.g. weeks). Planning depth H counts decision intervals, $t_{k+H} \le T^{\text{th}}$; it is not the observation count. Below k indexes decisions and $\mathcal{F}_k$ contains only released information. Freeze training length N, PCA rank $r < n$, scales, priors, calendar/surprise laws, risk/cost bounds, seed and validation splits. All money uses one currency. Do not roll the thesis deadline forward after losses.

Objects: symbol, type, dimensions	Financial meaning, estimator / equation, and downstream consumer	Python representation
F0. Data, gains, coordinates $O_{\le k}$: dated records; $g_s, x_s \in \mathbb{R}^n$. $D \in \mathbb{R}^{n\times n}$ positive diagonal; $\bar r \in \mathbb{R}^n$. $B = [V\ U] \in \mathbb{R}^{n\times n}$ orthogonal, $V \in \mathbb{R}^{n\times r}$.	Released settlements, multipliers m_i, liquidity, funding, margin and fills enter $g_{i,s} = m_i(F_{c_i(s-1),s} - F_{c_i(s-1),s-1})$, using the maturity held over that interval. Never splice a roll price jump into gain. On N training gains, $D_{ii} = \mathrm{sd}(g_i) > 0$, $r_s = D^{-1}g_s$, $\bar r = N^{-1}\sum r_s$. Eigenvectors of $S_r = (N-1)^{-1}\sum(r_s - \bar r)(r_s - \bar r)^\top$ give B, ordered with fixed signs/ties. Fast factor/residual coordinates: $x_s = B^\top(r_s - \bar r)$. Freeze/recompute the whole block in one basis. **To F1–F3, F5.**	NumPy data `(N,n)`, basis/covariance `(n,n)`, state `(n,)`; timestamped contract records.
F1. One-step baseline fit $\widehat A \in \mathbb{R}^{n\times n}$, $\widehat Q \in \mathbb{S}^n_{++}$; $\mu_k \in \mathbb{R}^n$, $\Sigma_k \in \mathbb{S}^n_{++}$. $\widehat\theta$: all fitted coefficients.	$x_{s+1} = Ax_s + \varepsilon_{s+1}$, $\varepsilon \sim \mathcal{N}(0, Q)$. $\widehat A = (\sum x_{s+1}x_s^\top)(\sum x_s x_s^\top + \lambda I)^{-1}$, $\widehat Q = (N-1)^{-1}\sum(x_{s+1} - \widehat A x_s)(x_{s+1} - \widehat A x_s)^\top + \epsilon I$, $\lambda, \epsilon > 0$. Reject spectral radius $\rho(\widehat A) \ge r_* < 1$. Baseline $g_{k+1} \mid \mathcal{F}_k \sim \mathcal{N}(\mu_k, \Sigma_k)$, $\mu_k = D(\bar r + B\widehat A x_k)$, $\Sigma_k = DB\widehat Q B^\top D$. **To F3, F5; this is one step, not a thesis-horizon forecast.**	Arrays `(n,n)`, `(n,)`; fit/transition callables; SciPy Gaussian law.
F2. Slow thesis extension $u_k \in \mathbb{R}$: latent valuation anchor; $\ell_k \in \mathbb{R}$: observed reference-spread level. $d_k = u_k - \ell_k$: remaining gap; $y_k \in \mathbb{R}$: economic proxy. $c_{k+1} \in \mathbb{R}^p$: catalyst surprises; $b_k = \mathrm{Law}(u_k \mid \mathcal{F}_k)$.	Choose a fixed reference basket $w_0 \in \mathbb{R}^n$, capital $A_0 > 0$, and a predeclared economic valuation proxy y in the same units: $\ell_{k+1} = \ell_k + w_0^\top g_{k+1}/A_0$, $u_{k+1} = \bar u + \phi(u_k - \bar u) + \beta^\top c_{k+1} + \sigma_u\eta_{k+1}$, $y_k = u_k + \tau_y\nu_k$, $x_{k+1} = Ax_k + a_d d_k + J_c c_{k+1} + \varepsilon_{k+1}$. Here $a_d \in \mathbb{R}^n, J_c \in \mathbb{R}^{n\times p}, \beta \in \mathbb{R}^p$, $0 < \phi \le 1$, $\sigma_u, \tau_y > 0$; independent standardized Gaussian η, ν. Persistence is estimated ϕ near one and small σ_u, not immunity to evidence. At announced releases $c \sim \mathcal{N}(\bar c, C_c)$, $C_c \in \mathbb{S}^p_+$, fitted to past surprises; otherwise $c = 0$. Filter from a proper prior: $b_{k+1}(u) \propto p(y_{k+1} \mid u)\int p(x_{k+1}, u \mid x_k, u', \ell_k) b_k(u')\,\mathrm{d}u'$; maximize predictive log likelihood (EM) for coefficients. Between decision-grid VAR observations, released y's update the current anchor likelihood; quotes update risk. $a_d = J_c = 0$ recovers F1. **To F3–F6.**	Filter mean/variance `(1,)`, `(1,1)`; coefficients `(n,p)`, `(p,)`. Proxy map and catalyst calendar fixed before testing.
F3. Joint physical path law $Z_k = (x_k, u_k, \ell_k) \in \mathbb{R}^{n+2}$; K^P_θ: F1/F2 transition. $\mathsf{P}^H_k \in \mathcal{P}((\mathbb{R}^{n+2})^{H+1})$. $L_k \in \mathbb{R}^{n\times n}$, $\rho_\mu \ge 0$: mean uncertainty.	$\mathsf{P}^H_k(\mathrm{d}Z_{k:k+H}) = b_k(\mathrm{d}u_k)\prod_{j=k}^{k+H-1} K^P_{\widehat\theta}(\mathrm{d}Z_{j+1} \mid Z_j)$, observed x_k, ℓ_k fixed, catalysts integrated. The endpoint $\mathbb{P}(Z_{k+H} \mid \mathcal{F}_k)$ is only a marginal. For F1 alone it gives $x_{k+H} \sim \mathcal{N}(A^H x_k, \sum_{j=0}^{H-1} A^j Q (A^j)^\top)$; cumulative gain and interim risk need cross-time dependence. Use $\mu_k = \mathbb{E}^P_k g_{k+1}$, $\Sigma_k = \mathrm{Cov}^P_k(g_{k+1})$, including anchor uncertainty. Block-bootstrap refits give $\mu_k^{(b)}$, $L_k L_k^\top = \widehat{\mathrm{Cov}}_b(\mu_k^{(b)}) + \epsilon_\mu I$; ρ_μ is a frozen quantile of $\|L_k^{-1}(\mu_k^{(b)} - \mu_k)\|$. Hold each fitted parameter set fixed per path. **To F5–F6.**	Paths `(N_paths,H+1,n+2)`, gains `(N_paths,H,n)`; Gaussian distribution or simulation callable; bootstrap means `(N_boot,n)`.
F4. Thesis status / expression $p_k = b_k(u_k - \ell_k > \delta) \in [0,1]$; $z_k \in \{0, 1, \dagger\}$: waiting, active, closed. $p_{\text{on}} > p_{\text{off}}, \delta, \delta_{\text{exit}} \ge 0$; eligible mask $j_k \in \{0,1\}^n$.	For a positive-gap thesis, activate at $p_k \ge p_{\text{on}}$; retain it between thresholds. Close on convergence $\mathbb{E}_k d_k \le \delta_{\text{exit}}$, $p_k \le p_{\text{off}}$, released adverse evidence $y < y_{\min}$ or $c_j < \underline{c}_j$, or T^{th}; freeze thresholds and label reasons. Waiting forbids new thesis risk; † cancels entry orders and flattens, and is absorbing until a new thesis record. Risk/margin failures act immediately. The thesis is u, d, b, not a contract ID: choose maturities, spreads, hedges or zero anew. Shares/options require their own pricing/ledger. Eligibility $j_i = 1$ requires valid liquid quotes, capped spreads and dates before notice/roll cutoff; held ineligible legs exit. Disabling this extension uses F1/F5 without the thesis gate. **To F5–F6.**	Scalar probabilities, mode/deadline record, Boolean mask `(n,)`. Log thesis exit separately from temporary zero exposure.

Futures: strategic control, execution and accounting

The baseline uses H=1. Longer planning adds continuation value; observing another tick does not itself authorize another position change.

Objects: symbol, type, dimensions	Financial meaning, estimator / equation, and downstream consumer	Python representation
F5. Baseline target / constraints $s^{\text{acct}} = (q^-, X, M) \in \mathbb{Z}^n \times \mathbb{R}^2$, $W = X + M$. Target $q \in \mathbb{Z}^n$, roll map $\mathcal{R}(q) \in \mathbb{Z}^{2n}$. $s, \eta \in \mathbb{R}^{2n}_+$, $\ell^M \in \mathbb{R}^n_+$: spread/impact/margin.	With diagonal roll flags R^{roll}, $a = \mathcal{R}(q) = (-R^{\text{roll}}q^-, q - (I - R^{\text{roll}})q^-)$. Read spreads/margin; fit nonnegative impact from fills. Define $K(a) = s^\top\lvert a\rvert + \sum \eta_i a_i^2$, $M^+(q) = (\ell^M)^\top\lvert q\rvert$, $d^c(q) = (R^M - R^f)M^+(q) - R^f K(a)$, $v(q) = q^\top \Sigma_k q$. Baseline score $J_1(q) = [q^\top\mu_k - \rho_\mu\|L_k^\top q\| + d^c(q)]/A_0 - \gamma v(q)/(2A_0^2)$. Feasible C_k: $\lvert q_i\rvert \le j_i \bar{q}_i$, $\|Dq\|_1 \le L_g$, $\lvert V^\top Dq\rvert \le b$, $\lvert H_g Dq\rvert \le c_g$, $W - K - M^+ \ge R_{\text{cash}}$, stress-equity floors, and $-q^\top\mu + \rho_\mu\|L^\top q\| - d^c + \sqrt{v}\,\varphi(\Phi^{-1}(\alpha))/(1-\alpha) \le L_{\text{ES}}$. This Gaussian ES formula applies only to the one-step Gaussian law; use simulated loss ES for longer paths. **To F6.**	CVXPY integer vector `(n,)` or bounded enumeration; constraints use supplied caps, $H_g \in \mathbb{R}^{p_g \times n}$. Gross factors $R^f, R^M > 0$; normal PDF/CDF φ, Φ.
F6. Bellman / stochastic MPC $s_k = (\mathcal{F}_k$ summary, $b_k, z_k, T^{\text{th}} - t_k, s_k^{\text{acct}}$, pending orders). $V^h, Q^h : \mathcal{S} \to \mathbb{R}$, $\mathcal{S} \times \mathcal{A} \to \mathbb{R}$. $\eta_T, h_0 \ge 0$: turnover/switch preferences.	Let $\Pi = W' - R^f W$ from F7 and $r_F = \Pi/A_0 - \gamma\,\mathrm{Var}_k(\Pi)/(2A_0^2) - \rho_\mu\|L_k^\top q\|/A_0 - \eta_T\|a\|_1$. $Q_k^h(s,a) = \mathbb{E}_k^P[r_F + V_{k+1}^{h-1}(s')]$, $V_k^h(s) = \max_{a \in \mathcal{A}_k(s)}\{Q_k^h(s,a) - h_0 \mathbf{1}_{a \ne 0}\}$. Choose $V^0 = 0$ to recover F5 at $H = 1, \eta_T = h_0 = 0$; use a specified continuation value at a shorter planning boundary, and $-K^{\text{liq}}/A_0$ at the thesis endpoint. Simulate proxy observations and re-filter at future nodes: actions never see latent states or later shocks. Stochastic MPC executes the first action of this nonanticipative tree; full Bellman control uses the remaining horizon. No-action region: $\max_{a \ne 0}[Q^H(s,a) - Q^H(s,0)] \le h_0$, if holding is feasible. F4 hysteresis and this band suppress reversals; mandatory exits/rolls override them. **To F7.**	Scenario tree / value callable; finite belief representation $\varphi_s(s) \in \mathbb{R}^{d_s}$. Costs are already in Π; turnover and switch penalties are preferences, not second cash debits.
F7. Separate execution / ledger $a^{\text{str}} \in \mathbb{Z}^{2n}$: strategic parent order; $y_e, v_e \in \mathbb{R}^{2n}$: remaining order/rate. $\Lambda_e \succ 0, \Omega_e \succeq 0 \in \mathbb{R}^{2n \times 2n}$, $T_e \ll \Delta_d$, $\lambda_e \ge 0$.	Execution chooses $v_e = -\dot{y}_e$ to minimize $\int_0^{T_e}[s_e^\top\lvert v_e\rvert + v_e^\top\Lambda_e v_e + \lambda_e y_e^\top\Omega_e y_e]\,\mathrm{d}u$, $y_e(0) = a^{\text{str}}$, $y_e(T_e) = 0$, subject to lot/rate/interim-risk limits. Fit impact Λ_e from fills and quote-change covariance Ω_e; carry unfilled quantities if completion fails. Feed expected cash cost back into F5/F6, replacing its cost estimate; execution-risk penalty is separate. Full immediate fills remain the baseline: $q^+ = q^- + \tilde{a}^{\text{old}} + \tilde{a}^{\text{new}}$, $X^+ = X + M - M^+(q^+) - K(\tilde{a})$, $W' = R^f X^+ + R^M M^+(q^+) + (q^+)^\top g'$, $\Pi = W' - R^f W$. Reset $M' = (\ell^{M'})^\top\lvert q^+\rvert$, $X' = W' - M'$. With gradual fills, accrue cash event by event and use $\int q(u)^\top\,\mathrm{diag}(m)\,\mathrm{d}F(u)$ over actual maturities. **To next account/filter.**	Execution optimizer/callable; order/fill arrays `(2*n,)`; event ledger. Strategic controller chooses exposure; execution chooses timing/size of child orders.
F8. Optional RL; when / interface $Q_\psi : \mathbb{R}^{d_s+2n} \to \mathbb{R}$: learned action value; $\bar{\psi}$: target parameters.	Input the full finite belief/account/mode/deadline encoding and parent action. Train from supported transitions (s, a, r_F, s') on $\mathcal{L}(\psi) = \mathbb{E}[(Q_\psi(s,a) - \{r_F - h_0\mathbf{1}_{a \ne 0} + \max_{a' \in \mathcal{A}(s')} Q_{\bar{\psi}}(s', a')\})^2]$, replacing the maximum by terminal value at the endpoint. Use RL when repeated large control solves need approximation, or action-dependent fills/information must be learned from adequate data. It is unnecessary when the fitted law and constrained MPC/Bellman solve are tractable. Validate simulator/offline coverage and preserve hard constraints; RL cannot supply an unidentified thesis edge. **Replaces F6 value solve only.**	Model callable / tensors; sufficient belief encoding (not an arbitrary posterior mean), feasible-action mask, replay transitions. Unseen actions require evidence, not extrapolation claims.
F9. Validation / gates $d_j^{\text{test}} = (\Pi_j - \Pi_j^{\text{bench}})/A_0$; J test decisions; $\widehat{\Gamma}_l$: sample lag covariance.	Chronological train/select/test blocks; test joint-path calibration, thesis persistence, catalyst forecasts, turnover, execution and net gains against frozen benchmarks. $Z = \sqrt{J}(\bar{d}^{\text{test}} - \delta_{\text{econ}})/\sqrt{\widehat{\Omega}}$, $\widehat{\Omega} = \widehat{\Gamma}_0 + 2\sum_{l=1}^{L_v}(1 - l/(L_v+1))\widehat{\Gamma}_l$. Predeclare thresholds/multiplicity; resample blocks preserving overlap and longer observed dependence. Gate solves only for valid inputs and nonempty feasible sets; otherwise funded liquidation or default/stop. Save posterior, gap, value difference and gate reason. **To next F0 and action gate.**	Scores `(J,)`, path diagnostics, block resampling; mode/risk/model/execution exits have distinct reason codes.

Composition: $O_{\le k} \to (x_k, \ell_k, y_k) \to (\widehat{\theta}, b_k, d_k, z_k) \to \mathsf{P}_k^H \to (Q_k^H, \mathcal{A}_k) \to a_k^{\text{str}} \to$ child fills $\to W_{k+1} \to b_{k+1}$.

$$a_k^* = F(O_{\le k}, s_k^{\text{acct}}, \widehat{\theta}) = \mathcal{E}_{\text{F7}}\left[\text{Gate}_{\text{F4,F9}}\left(\arg\max_{a \in \mathcal{A}_k(s_k)}\left\{\mathbb{E}_{\mathsf{P}_k^H}[r_F(s_k, a, s') + V_{k+1}^{H-1}(s')] - h_0\mathbf{1}_{a \ne 0}\right\}\right)\right].$$